

10 OPTIONS FOR CHECKWEIGHER

This chapter describes the function of the options for checkweigher.

TIP

- The function described in this chapter is not equipped for standard model. To use these functions, Purchasing the software license key or adding/converting hardware is required.

10.1 Rollover Prevention Board

The resin board (Rollover prevention board) is mounted on a checker with arm rejecter to prevent products from exceeding beyond the arm or falling when rejecting products.

This board is effective in preventing the product that is relatively high and tends to fall easily.

NOTE

- To prevent the machine damage, the arm speed must be set below the medium speed.
- "Fig. 10-1 " shows the rejecter device of SS specification.

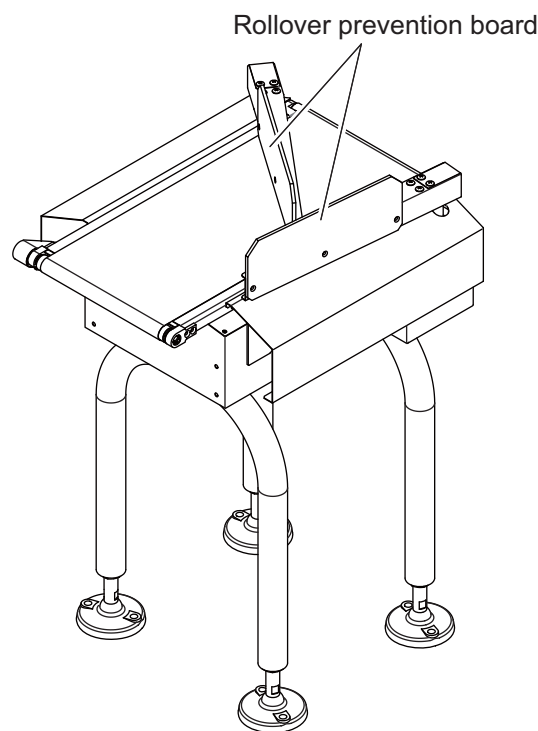


Fig. 10-1

10.2 Signal Tower Setting

< Overview of the Signal Tower Setting Item >

The items and function of Signal Tower Setting is as follows. Regarding the setting method, refer to the procedures in the chapter mentioned in the reference column below.

Table 10-1

Item	Function	Reference
Operation*	Sets the signal to be output during operation.	"10.2.1 Output Condition Setting (Signal Tower)"
Conveyor-running*	Sets the signal to be output while the conveyor is running.	
Stop*	Sets the signal to be output during stoppage.	
Error Stop*	Sets the signal to be output during error stoppage.	
Proper Wt.	Sets the signal to be output when a proper item passes.	
Over Wt.	Sets the signal to be output when an over weight signal passes.	
Under Wt.	Sets the signal to be output when an under weight signal passes.	
OK-over Wt.	Sets the signal to be output when the "OK-over" item passes.	
OK-under Wt.	Sets the signal to be output when the "OK-under" item passes.	
Metal Detection	Sets the signal to be output when a metal-containing item passes.	
Ext. 1-2	Sets the signal to be output for the item to be rejected in the set direction when a signal is input from External unit 1.	
Pitch Error	Sets the signal to be output for the item to be rejected in the set direction when a pitch error occurs.	
Frn Obj.	Sets the signal to be output for the item to be rejected in the set direction when a foreign object error occurs.	
Product Length Error	Sets the signal to be output for the item to be rejected in the set direction when a product length error occurs.	

10.2.1 Output Condition Setting (Signal Tower)

Output Condition Setting (Signal Tower) menu defines which signal tower should be lit corresponding to each condition such as overweight and under operation.
Output Condition Setting menu (Signal Tower) is as follows.

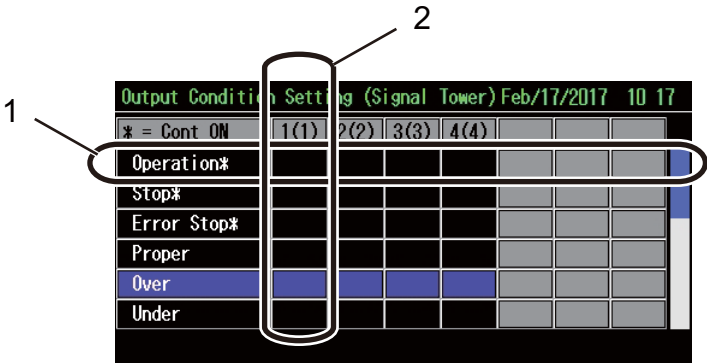


Fig. 10-2

Table 10-2

No.	Description
1	The signal tower setting status for each condition is displayed. -V mark: Set -Blank: Not set - - mark: Setting impossible (Conveyor synchronization, Under operation, and others have been set.)
2	Each setting state of the signal tower is displayed.

To set the Output Condition Setting (Signal Tower), follow the procedure below.

1. Display the Setup menu screen.
2. Select and enter the Signal Tower Setting.
▶ The Signal Tower Setting menu is displayed.

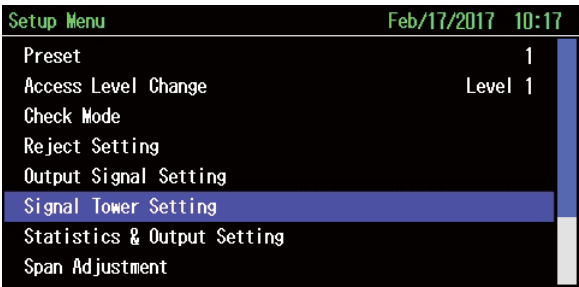


Fig. 10-3

3. Select and enter the output condition setting.
▶ The warning screen is displayed.

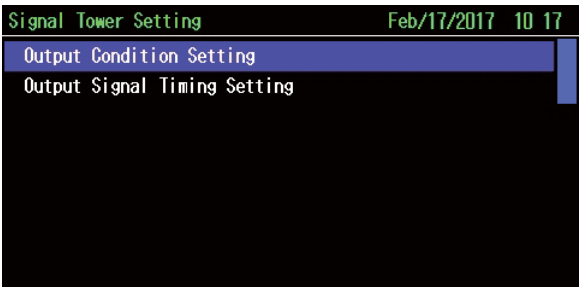


Fig. 10-4

4. Press the [Enter] key.

► Output Condition Setting menu (signal tower) screen is displayed. All signals currently output are turned off.

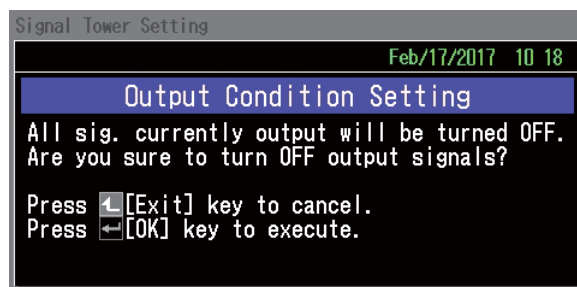


Fig. 10-5

5. Select and enter the preferred conditions with the [Up] or [Down] key.

► Output Condition Setting menu (signal tower) screen is displayed.

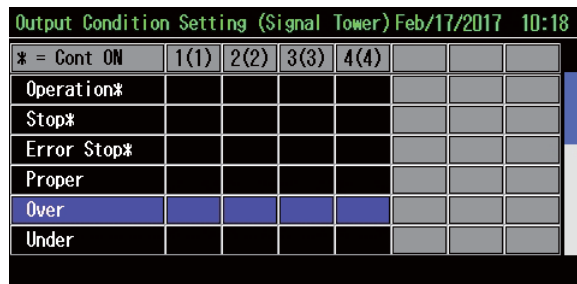


Fig. 10-6

6. Press the [Enter] key when the warning screen is displayed.

NOTE

- The warning screen is displayed when under operation or conveyor synchronization.

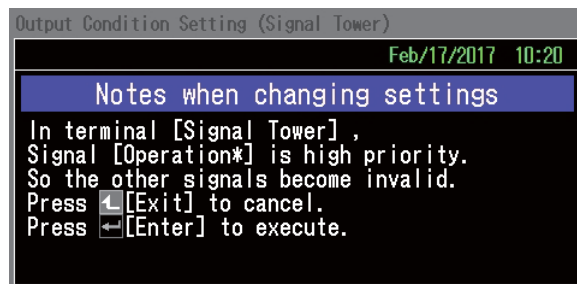


Fig. 10-7

7. Set "ON" or "OFF", then press the [Exit] key.

8. Press the [Exit] key after completing the setting.

► Stopped signal is output.

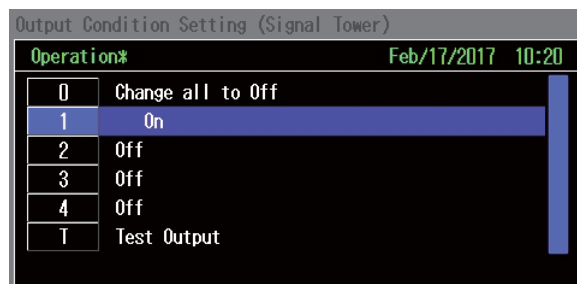


Fig. 10-8

9. When selecting "Test Output", signals are output to all signal towers.

10. Press any key to cancel the output.

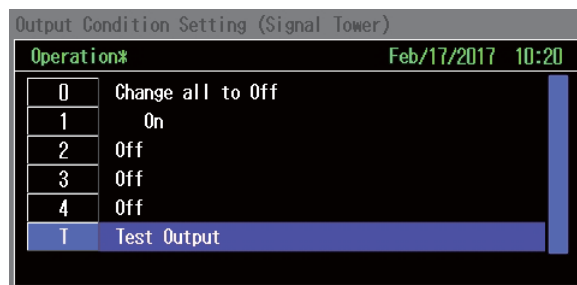


Fig. 10-9

TIP

- Short cut keys are follows.
[1] key: "ON/OFF" of the selected line 1
[2] key: "ON/OFF" of the selected line 2
[3] key: "ON/OFF" of the selected line 3
[4] key: "ON/OFF" of the selected line 4
[C] key: Sets "ON/OFF" to all signal tower of the selected line.

Output Condition Setting (Signal Tower) Feb/17/2017 10 20							
* = Cont ON	1(1)	2(2)	3(3)	4(4)			
Operation%	✓						
Stop%							
Error Stop%							
Proper	—						
Over	—						
Under	—						

Fig. 10-10

10.2.2 Output Timing Setting (Signal Tower)

In Output Timing Setting (Signal Tower), the output delay time, ON Time, and flashing interval can be set for each signal tower. Output Timing Setting menu screen (Signal Tower) is as follows.

Output Timing Setting (Signal Tower) Feb/17/2017 10 24					
Sig	NC/NO/COM	DelayTim(1)	ON Time (2)	FlkIntvl(3)	
1		5000 ms	Cont. ON	500 ms	
2		0 ms	1000 ms	0 ms	
3		0 ms	1000 ms	0 ms	
4		0 ms	1000 ms	0 ms	

Fig. 10-11

Table 10-3

No.	Description
1	Displays the timing setting state for each signal tower. Select the preferred signal tower with the [Up] or [Down] key.
2	Displays the delay time for each signal tower.
3	Displays the ON Time for each signal tower.
4	Displays the flashing interval for each signal tower.

NOTE

- When the ON Time is displayed as "Continuous ON" in red, the signal tower has been set as a signal of "under operation" or "conveyor synchronization". They can be set only "Continuous ON" so that signals will not be overlapped.
- When "Continuous ON" displayed at the Output Signal ON Time is alternately flashing in red and white, the port has been set as a signal of "Stopped*" or "Error Stop*". Although "Stopped*" or "Error Stop*" is output as "Continuous ON", they can be overlapped with other signals. Other signals are output in Output Signal ON Time displayed in white.

For the output timing setting, follow the procedures below.

1. Display the Setup menu screen.
2. Select and enter the signal tower setting.

► The Signal Tower Setting screen is displayed.

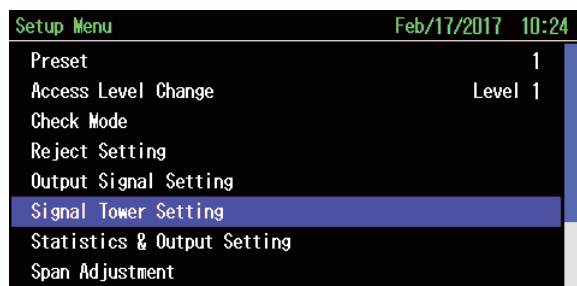


Fig. 10-12

3. Select and enter the output signal timing setting.

► The Output Timing Setting (Signal Tower) menu is displayed.

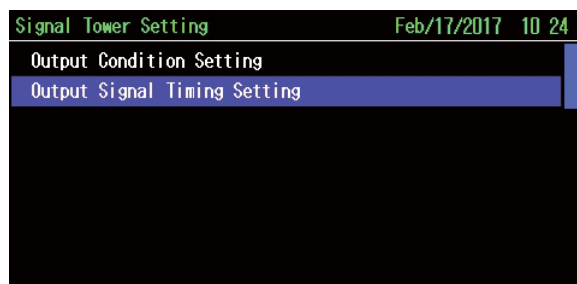


Fig. 10-13

4. Select and enter the signal tower.

► The signal menu screen is displayed.

Output Timing Setting (Signal Tower) Feb/17/2017 10:25				
Sig.	NC/NO/COM	DelayTim(1)	ON Time (2)	BlkIntvl (3)
1		5000 ms	Cont. ON	500 ms
2		0 ms	1000 ms	0 ms
3		0 ms	1000 ms	0 ms
4		0 ms	1000 ms	0 ms

Fig. 10-14

5. Select the item.

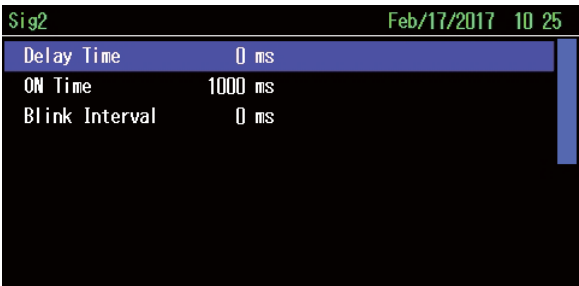


Fig. 10-15

6. Input and enter numerical value.

7. Press the [Exit] key after completing the setting.

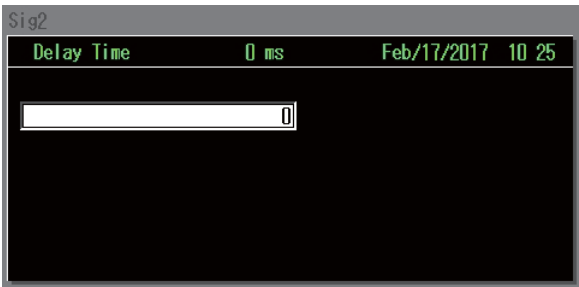


Fig. 10-16

TIP

- Short cut keys are follows.
[1] key: "Delay time" of the selected line.
[2] key: "Output Signal ON Time" of the selected line.
[3] key: "Flashing interval" of the selected line.

Output Timing Setting (Signal Tower) Feb/17/2017 10:25				
Sig.	NC/NO/COM	DelayTim(1)	ON Time (2)	BlkIntvl(3)
1		5000 ms	Cont. ON	500 ms
2		0 ms	1000 ms	0 ms
3		0 ms	1000 ms	0 ms
4		0 ms	1000 ms	0 ms

Fig. 10-17

10.3 Proper Weight Count

Proper Weight Count counts the number of "proper product" and rejects the status of the proper product into up to 3 patterns such as "proper product 1", "proper product 2", and "proper product 3" based on the Proper Weight Count.

This item can also change the Proper Wt. Rej. Dir. and output a contact signal based on the number of the proper product.

NOTE

- To use the Proper Weight Count, "License key" is required.
- When the weight zone is set to "5 zone" in the system configuration at the installation level, the "Proper Weight Count" menu is not displayed.

10.3.1 Total Count Zoom Display

10.3.1.1 Overview

Total Count Zoom display (Standby menu) shows the "Prop1.LotCount", "Prop2.LotCount", "Prop3.LotCount", "Prop.Ttl LotCnt", "Prop1. Wt. Count", "Prop2. Wt. Count", and "Prop3. Wt. Count", and counts when a proper product passes.

The lot number is the value divided proper product by proper weight count. For example, "Prop1.LotCount" is divided "Prop1. Wt. Count" by "Proper Weight Count".

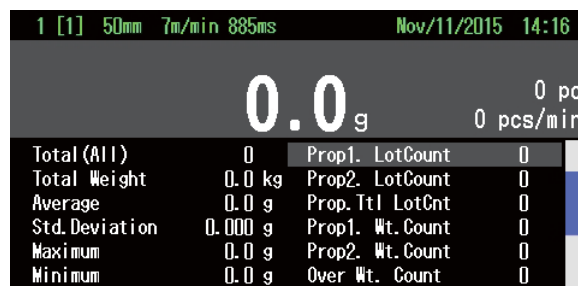


Fig. 10-18

NOTE

- In addition to standard clear operation, the total count result will also be cleared in Total Count Zoom in the following cases.
 - a. When changing the proper weight count and pressing the [Start] key.
 - b. When changing the Proper Reject Setting and pressing the [Start] key.

10.3.1.2 Proper Count Clear

Proper Count Clear clears the current proper weight count including the proper weights 1, 2, and 3. After clearing them, it starts from "Proper weight 1".

NOTE

- Proper Count Clear should be set by the site engineer or higher level personnel.
- The count number is not cleared only when the operation stops. The count number is migrated when starting the operation next time.

To clear the Proper Weight count, follow the procedure below.

1. Press the [Setup] key in the Standby menu.

► The Setup menu is displayed.

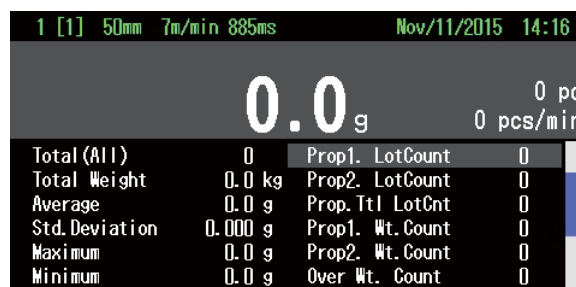


Fig. 10-19

2. Select and enter the "Proper Count Clear".

► The Confirm screen is displayed.

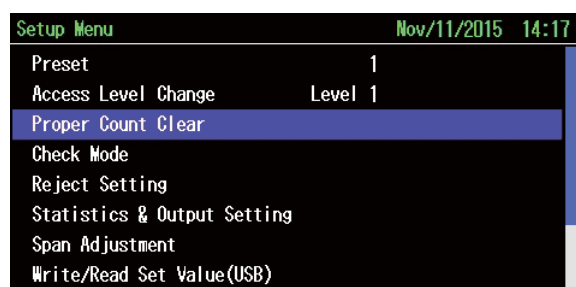


Fig. 10-20

3. Press the [Enter] key.

► The Setup menu is displayed.

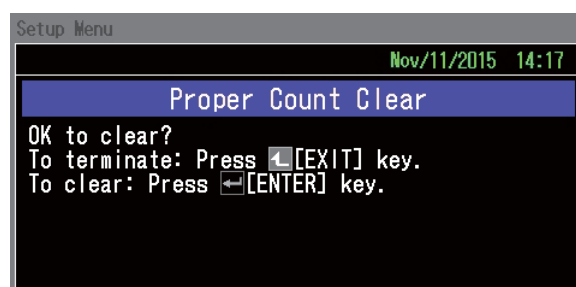


Fig. 10-21

NOTE

- In addition to the Proper Count Clear, the Proper Weight Count will also be cleared in the following cases.
 - a. When power is turned on.
 - b. When the proper weight count is reached.
 - c. When changing the preset setting and press the [Start] key.
 - d. When changing the proper weight count and press the [Start] key.

10.3.2 Basic Setting

NOTE

- Basic Setting should be set by the site engineer or higher level personnel.

10.3.2.1 Proper Count

Proper Count sets the proper product count. When the number of the proper product reaches the set count number, the reject direction will be switched, for example, from "proper product 1" to "proper product 2". To set the Proper Count, follow the procedure below.

- Press the [Preset] key in the Standby menu.

- ▶ The Preset Number Select Screen is displayed.

The screen displays the following information:

1 [1] 50mm 7m/min 885ms		Nov/11/2015 14:17	
0.2g		0 pc	
		0 pcs/min	
Total (All)	0	Prop1. LotCount	0
Total Weight	0.0 kg	Prop2. LotCount	0
Average	0.0 g	Prop. Ttl LotCnt	0
Std. Deviation	0.000 g	Prop1. Wt. Count	0
Maximum	0.0 g	Prop2. Wt. Count	0
Minimum	0.0 g	Over Wt. Count	0

Fig. 10-22

- Select and enter the desired preset number.

- ▶ The Preset menu (Basic) screen is displayed.

The screen displays the following information:

Preset Number		Nov/11/2015 14:17	
Preset Number	1		
1	0.0 [1] 50mm 7m/min 885ms		
2	0.0 [0] 50mm 7m/min 885ms		
3	0.0 [1] 235mm 60m/min 210ms		
4	0.0 [1] 195mm 40m/min 360ms		
5	0.0 [0] 195mm 40m/min 360ms		
6	0.0		
Press [ENTER] key to set. Simple Preset			

Fig. 10-23

- Select and enter the "Detailed Setting".

- ▶ The Preset menu (Detailed) screen is displayed.

The screen displays the following information:

Preset		Nov/11/2015 14:18	
Preset Number	1		
Upper Limit+	0.0 g		
Lower Limit-	0.0 g		
Product Length	50 mm		
Speed	12 pcs/min		
Conv. Speed	7 m/min		
← Detailed Setting			
		<	>

Fig. 10-24

- 4. Select and enter the "Proper Count".

►The Proper Count screen is displayed.

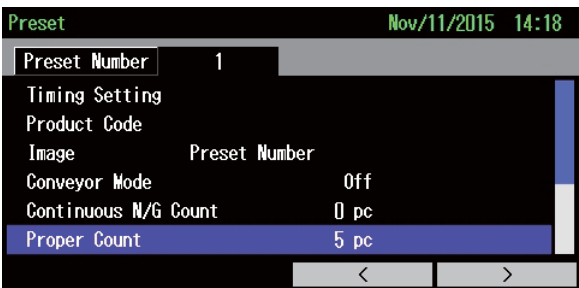


Fig. 10-25

- 5. Input and enter the numeric value.

►The Preset menu (Detailed) screen for the Proper Count setting will be displayed.

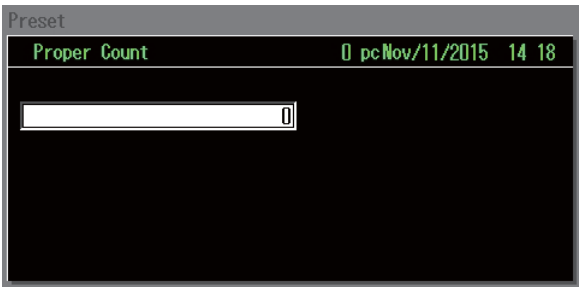


Fig. 10-26

10.3.2.2 Proper Reject Setting

Proper Reject Setting sets the direction number to reject the proper product. "1 direction" is for "Proper weight" (standard specification) only, "2 directions" is for "Proper weight 1" and "Proper weight 2" directions. Up to 3 directions can be set. To set the Proper Reject Setting, follow the procedure below.

- 1. Display the Preset menu (Basic) screen.

- 2. Select and enter the "Detailed Setting".

►The Preset menu (Detailed) screen is displayed.

- 3. Select and enter the "Proper Reject Setting".

►The Proper Reject Setting Setting screen is displayed.

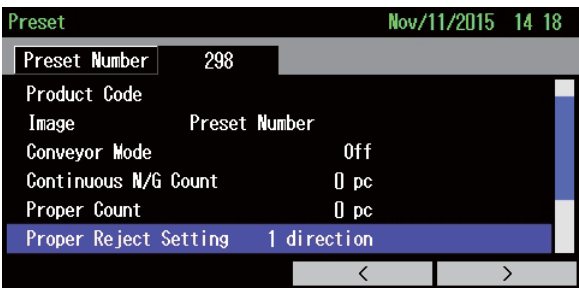


Fig. 10-27

- 4. Select and enter the directions of proper reject setting.

►The Preset menu (Detailed) screen for the Proper Reject setting will be displayed.

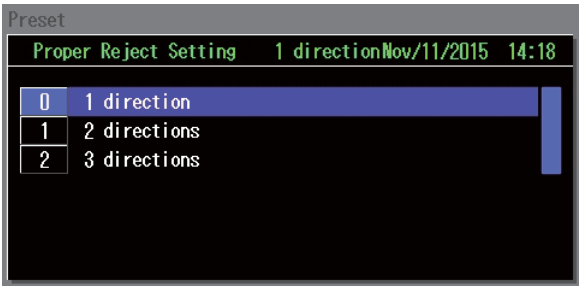


Fig. 10-28

10.3.2.3 Reject target time in proper weight switchover

In this time (Reject target time in proper weight switchover) the machine forcibly rejects the product as "Uninspected" during the set time after switching the "Proper Wt. Rej. Dir.", for example, from "Proper weight 1" to "Proper weight 2". The necessity for this item depends on the switching time of the rejector.

NOTE

- The default Forced Rejection Time is 0 ms.

To set the reject target time in proper weight switchover, follow the procedure below.

1. Display the Preset menu (Basic) screen.
2. Select and enter the "Detailed Setting".
 - The Preset menu (Detailed) screen is displayed.
3. Select and enter the "Timing Setting".
 - The Timing Setting screen is displayed.

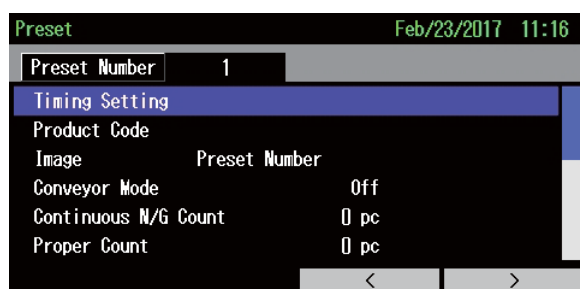


Fig. 10-29

4. Select and enter the "In P-wt Sw Rj Target Time".
 - The In P-wt Sw Rj Target Time screen is displayed.

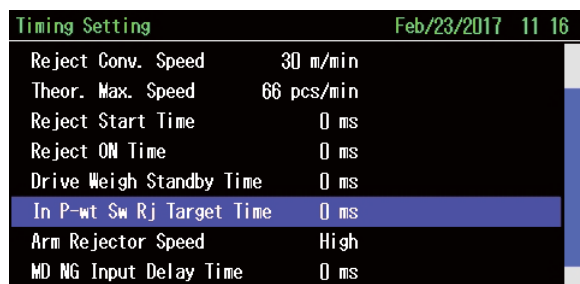


Fig. 10-30

5. Input and enter the numeric value.
 - The Timing Setting screen for the In P-wt Sw Rj Target Time setting will be displayed.

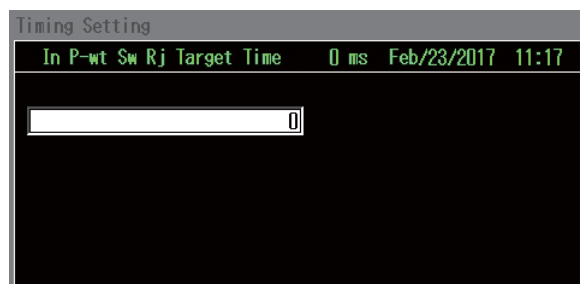


Fig. 10-31

10.3.3 Output Signal or Signal Tower Setting

This section explains the output signal setting or the signal tower setting with the specification of proper weight count.

<Overview of the Output Signal Condition Setting Item>

Table 10-4

Item	Function
Proper 1	Sets the signal to be output when the Proper 1 product passes through.
Proper 2	Sets the signal to be output when the Proper 2 product passes through.
Proper 3	Sets the signal to be output when the Proper 3 product passes through.
Proper 1 switching	Sets the signal to be output when starting the Proper 1 lot.
Proper 2 switching	Sets the signal to be output when starting the Proper 2 lot.
Proper 3 switching	Sets the signal to be output when starting the Proper 3 lot.
In proper weight switchover	Sets the signal to be output when the product passes through between the end of the current lot and the start of the next lot. (Refer to "10.3.2.3 Reject target time in proper weight switchover")

<Transition Diagram of Output Signal Status>

The figure below shows an example of signal status transition when setting the Output Condition, Output Signal on time, and Output Signal Delay time.



Fig. 10-32

< Example of setting output signal > When Interlocking to external rejection.

As a setting example, setting when interlocking to the rejecter of the following specification is described.

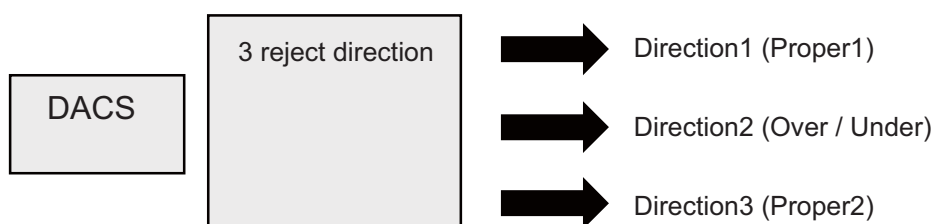


Fig. 10-33

The contact signal output of DACS and rejecter are wired as shown in "Table 10-5 ". The rejecter keep rejecting the product in the direction of inputting signal while inputted.

Table 10-5

Wiring for signal	reject direction
XT102-30, 32 (Contact a)	Direction 1 (Proper 1)
XT102-33, 35 (Contact a)	Direction 2 (Under / Over)
XT102-36, 38 (Contact a)	Direction 3 (Proper 2)

The setting value is as shown in "Table 10-6 ". See the "Reference chapter" for the setting procedure.

Table 10-6

Item	Setting value	Reference chapter
Proper Count	5 count	"10.3.2.1 Proper Count"
Proper Reject Setting	2 derrection	"10.3.2.2 Proper Reject Setting"
Proper1 Wt. Sig. (XT101)	32	"10.2.1 Output Condition Setting (Signal Tower)"
Proper2 Wt. Sig. (XT101)	38	
Over Wt. Sig. (XT101)	35	
Output Sig. 32 ON Time	0 ms	"10.2.2 Output Timing Setting (Signal Tower)"
Output Sig. 35 ON Time	0 ms	
Output Sig. 38 ON Time	0 ms	
Output Sig. 32 Delay (XT101)	0 ms	
Output Sig. 35 Delay (XT101)	0 ms	
Output Sig. 38 Delay (XT101)	0 ms	

< Example of setting signal tower > For Proper Weight Count

The setting when lighting the signal tower using the following specification is described.

The operator wants to check the status either "Proper1" or "Proper2" with the signal tower during production. Even a non-proper weight product (e.g. Over) comes in, not to light the signal tower to allow the status of proper weight to be checked all the time.

In addition, the buzzer (signal tower 1) sounds for one second to let the operator know the Proper weight is switched from Proper weight 1 to Proper weight 2.

The setting value is as shown in "Table 10-7 ". See the "Reference chapter" for the setting procedure.

Table 10-7

Item	Setting value	Reference chapter
Proper Count	5 count	"10.3.2.1 Proper Count"
Proper Reject Setting	2 derection	"10.3.2.2 Proper Reject Setting"
Proper1 Wt. Sig. (XT101)	1, 4	"10.3.3 Output Signal or Signal Tower Setting"
Proper2 Wt. Sig. (XT101)	1, 3	
Sig. Tower1 ON Time	1000 ms	
Sig. Tower3 ON Time	0 ms	
Sig. Tower4 ON Time	0 ms	

10.4 FBC INDIRECT FEEDBACK CONTROL (TREND OUTPUT)

This section provides information on the case DACS provides TREND OUTPUT with the auger filler and the auger filler control part controls the auger filler in conjunction with the work of the auger filler and DACS.

10.4.1 Overview of the auger filler feedback control

This equipment controls the auger filler so that filler output achieves the Target Weight with the following structure.

DACS-GN checks the weight of the filled packs and judges whether the quantity is excessive or insufficient using the average value of the weight. Based on this, the prescribed pulse count is increased or decreased to regulate auger filler output. Also this equipment outputs object sensing signal every time the filled packs go through.

The auger filler is controlled by the control part of the auger filler.

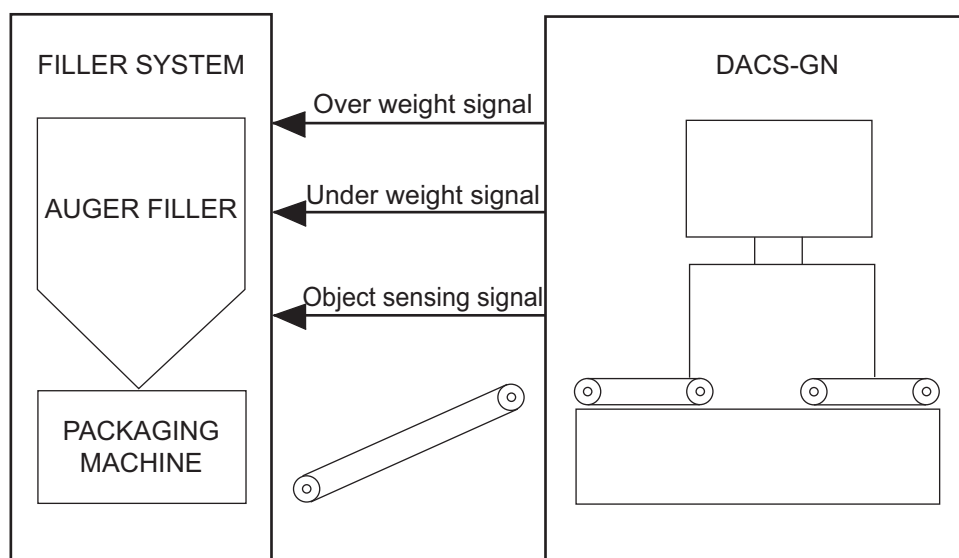


Fig. 10-34

10.4.2 Status Display

A display showing FBC status is added during the operation and standby. Pressing the [Display] key until the FBS status appears. Pressing the [Display] key each time switches the display. Follow the following procedure to display the FBC Status Display.

1. Press [Display] key during operation/standby.

► Pressing [Display] key each time switches the display in the following order:
"Weight Value Zoom Display"
(Fig. 10-35)



"Preset Information Display"
(Fig. 10-36)



"Histogram Display"
(Fig. 10-37)



"X-Bar Chart Display"
(Fig. 10-38)



"Last 20 Weight Data Display"
(Fig. 10-39)



"FBC READY STATUS SCREEN"
(Fig. 10-40)



"Pitch Adjustment Display"
(Fig. 10-41)



"Total Count Zoom Display"
(Fig. 10-42)

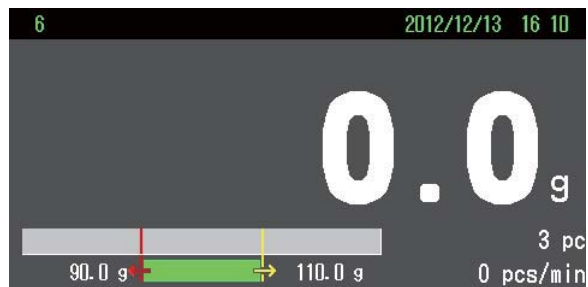


Fig. 10-35



Fig. 10-36



Fig. 10-37



Fig. 10-38

NOTE

- On "FBC READY STATUS SCREEN" ("Fig. 10-40 "), items are indicated below.

for Sample Count A/B
 B: The setting value of Sample Count
 A: The remaining value of Sample Count for FBC

for Sample Interval C/D
 D: The setting value of Sample Interval for FBC
 C: The remaining value of Sample Interval before sampling start for FBC
- On "FBC READY STATUS SCREEN" ("Fig. 10-40 "), select "ON"/"OFF" by [Up/Down] key to turn the FBC ON/OFF.



Fig. 10-39



Fig. 10-40



Fig. 10-41

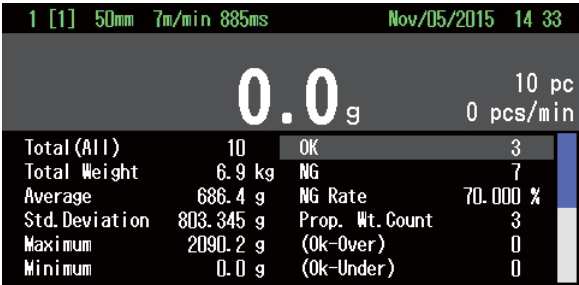


Fig. 10-42

10.4.3 Preset Parameters

An item for setting FBC is added to the Preset Setting. To access FBC Setting follow the procedures listed below.

NOTE

- Preset Setting is conducted with above level 1.

1. Display the Preset menu.
2. Select and enter the "Detailed Setting".
 - ▶ The Preset menu (Detailed) screen is displayed.
3. Select and enter the "FBC Setting".
 - ▶ The FBC Setting screen is displayed.

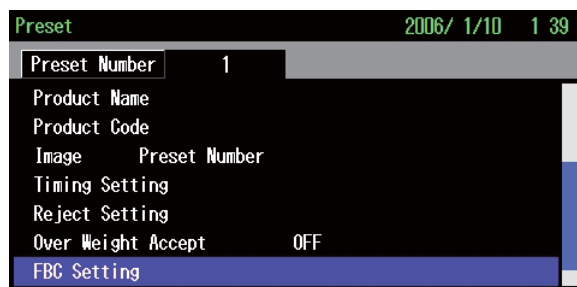


Fig. 10-43

10.4.3.1 FBC

This defines whether or not the equipment provides TREND OUTPUT based on weighing results. Setting "0" (OFF) does not provide TREND OUTPUT. Setting "1" (ON) provides TREND OUTPUT. To set FBC follow the procedures listed below.

1. Display the FBC Setting screen.
2. Select and enter the "FBC".

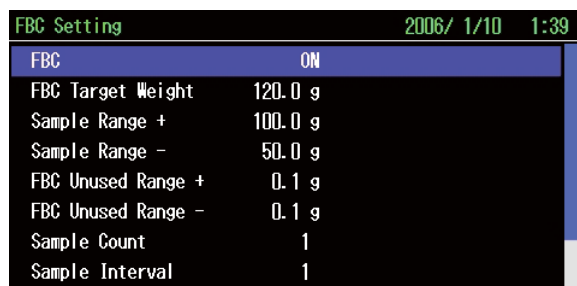


Fig. 10-44

3. Select and enter the "0" to deactivate Feedback Control. Select and enter the "1" to activate Feedback Control.
 - ▶ The FBC Setting screen is displayed.

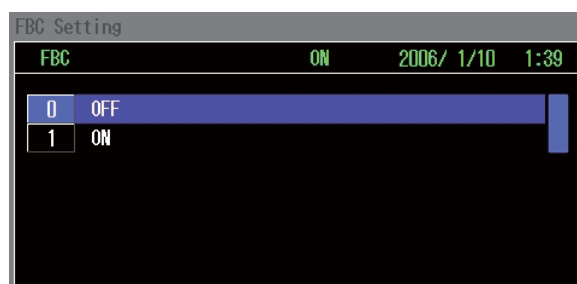


Fig. 10-45

10.4.3.2 FBC Target Weight

This defines the target net weight for packs filled by the auger filler.

To set FBC Target Weight follow the procedures listed below.

1. Display the FBC Setting screen.
2. Select and enter the "FBC Target Weight".

FBC Setting		2006/ 1/10	1:39
FBC	ON		
FBC Target Weight	120.0 g		
Sample Range +	100.0 g		
Sample Range -	50.0 g		
FBC Unused Range +	0.1 g		
FBC Unused Range -	0.1 g		
Sample Count	1		
Sample Interval	1		

Fig. 10-46

3. Input and enter the numeric value.
 - ▶ FBC Target Weight appears.
 - ▶ The setting for Target Weight is complete.

FBC Setting		2006/ 1/10	1:39
FBC Target Weight	120.0 g		
	0		

Fig. 10-47

10.4.3.3 Sample Range +

This defines the maximum pack weight above which the pack will not be included in the sampling.

To set Sampling Range + follow the procedures listed below.

1. Display the FBC Setting screen.
2. Select and enter the "Sampling Range +".

FBC Setting		2006/ 1/10	1:40
FBC	ON		
FBC Target Weight	120.0 g		
Sample Range +	100.0 g		
Sample Range -	50.0 g		
FBC Unused Range +	0.1 g		
FBC Unused Range -	0.1 g		
Sample Count	1		
Sample Interval	1		

Fig. 10-48

3. Input and enter the numeric value.

- ▶ Sampling Range + appears.
- ▶ The setting for Sampling Range + is complete.

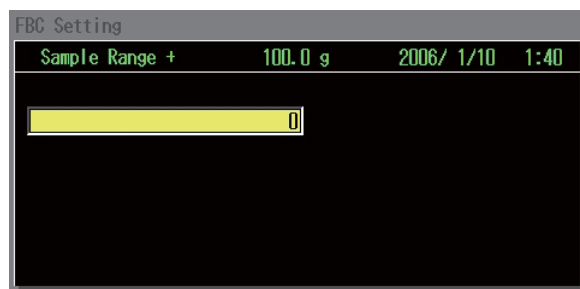


Fig. 10-49

10.4.3.4 Sample Range -

This defines the minimum pack weight below which the pack will not be included in the sampling. To set Sampling Range - follow the procedures listed below.

1. Display the FBC Setting screen.
2. Select and enter the "Sampling Range -".

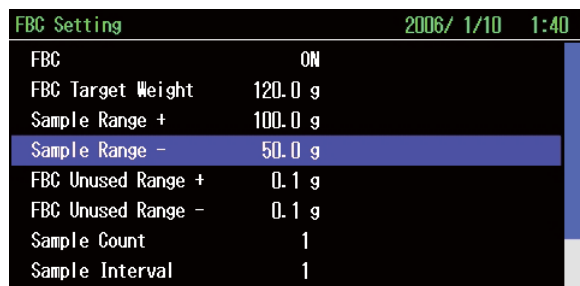


Fig. 10-50

3. Input and enter the numeric value.

- ▶ Sampling Range - appears.
- ▶ The setting for Sampling Range - is complete.

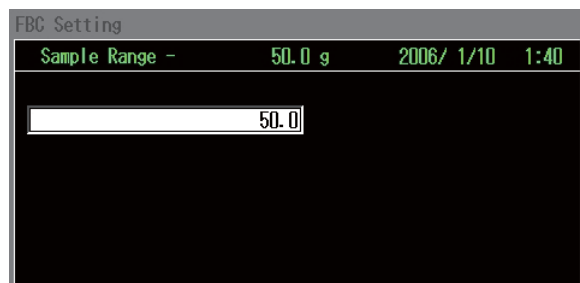


Fig. 10-51

10.4.3.5 FBC Unused Range +

This defines the average weight value which activates FBC and updates the control pulse when this weight is exceeded.

To set FBC Unused Range + follow the procedures listed below.

1. Display the FBC Setting screen.
2. Select and enter the "FBC Unused Range +".

FBC Setting		2006/ 1/10	1:40
FBC	ON		
FBC Target Weight	120.0 g		
Sample Range +	100.0 g		
Sample Range -	50.0 g		
FBC Unused Range +	0.1 g		
FBC Unused Range -	0.1 g		
Sample Count	1		
Sample Interval	1		

Fig. 10-52

3. Input and enter the numeric value.
 - ▶ FBC Unused Range + appears.
 - ▶ The setting for FBC Unused Range + is complete.

FBC Setting		2006/ 1/10	1:40
FBC Unused Range +	0.1 g		
	0.1		

Fig. 10-53

10.4.3.6 FBC Unused Range -

This defines the average weight value which activates FBC and updates the control pulse when this weight is not reached.

To set FBC Unused Range - follow the procedures listed below.

1. Display the FBC Setting screen.
2. Select and enter the "FBC Unused Range -".

FBC Setting		2006/ 1/10	1:40
FBC	ON		
FBC Target Weight	120.0 g		
Sample Range +	100.0 g		
Sample Range -	50.0 g		
FBC Unused Range +	0.1 g		
FBC Unused Range -	0.1 g		
Sample Count	1		
Sample Interval	1		

Fig. 10-54

3. Input and enter the numeric value.
 - ▶ FBC Unused Range - appears.
 - ▶ The setting for FBC Unused Range - is complete.

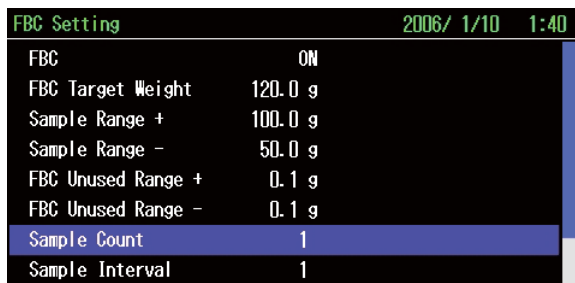
FBC Setting		2006/ 1/10	1:40
FBC Unused Range -	0.1 g		
	0.1		

Fig. 10-55

10.4.3.7 Sample Count

This defines the number of packs from which to derive the average value.
To set Sample Count follow the procedures listed below.

1. Display the FBC Setting screen.
2. Select and enter the "Sample Count".



FBC Setting		2006/ 1/10 1:40
FBC	ON	
FBC Target Weight	120.0 g	
Sample Range +	100.0 g	
Sample Range -	50.0 g	
FBC Unused Range +	0.1 g	
FBC Unused Range -	0.1 g	
Sample Count	1	
Sample Interval	1	

Fig. 10-56

3. Input and enter the numeric value.
 - ▶ Sample Count appears.
 - ▶ The setting for Sample Count is complete.

NOTE

- Set the value 1 or above.



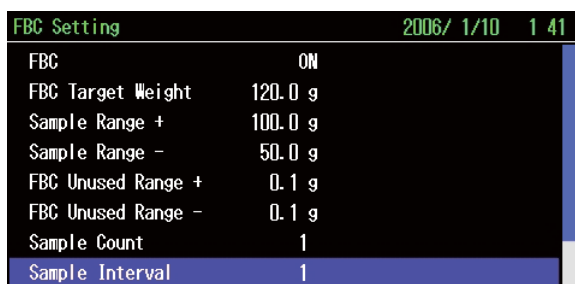
FBC Setting		2006/ 1/10 1:40
Sample Count	1	

Fig. 10-57

10.4.3.8 Sample Interval

This defines the interval from when the first pack using a new pulse count is sent until it is checked by DACS. Sampling is not taken during this interval.
To set Sample Interval follow the procedures listed below.

1. Display the FBC Setting screen.
2. Select and enter the "Sample Interval".



FBC Setting		2006/ 1/10 1:41
FBC	ON	
FBC Target Weight	120.0 g	
Sample Range +	100.0 g	
Sample Range -	50.0 g	
FBC Unused Range +	0.1 g	
FBC Unused Range -	0.1 g	
Sample Count	1	
Sample Interval	1	

Fig. 10-58

- 3. Input and enter the numeric value.
 - ▶Sample Interval appears.
 - ▶The setting for Sample Interval is complete.

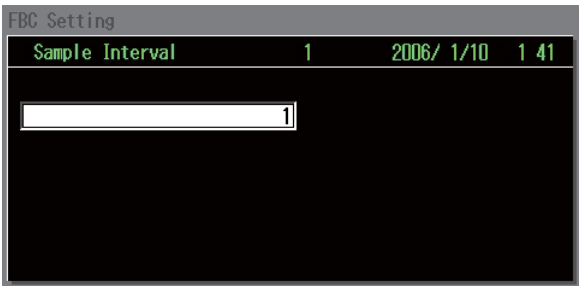


Fig. 10-59

10.4.4 System Configuration

An item is added to the System Configuration menu displaying FBC specifications. To set Gap Control follow the procedures listed below.

NOTE

- System Configuration is conducted with above level 2.

- 1. Display the System Configuration menu.
- 2. Select and enter the "Feedback Control".
 - ▶Feedback Control display appears.

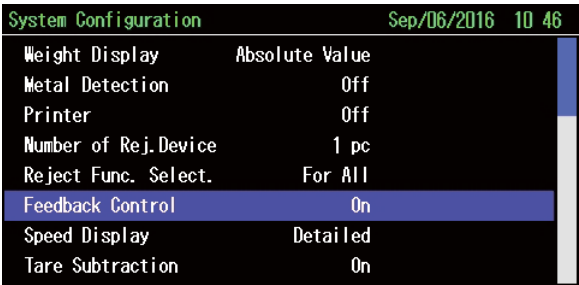


Fig. 10-60

- 3. Select and enter the "Off (0)" to deactivate Feedback Control.
Select and enter the "On (1)" to activate Feedback Control.
 - ▶System Configuration display in which Feedback Control is set appears.

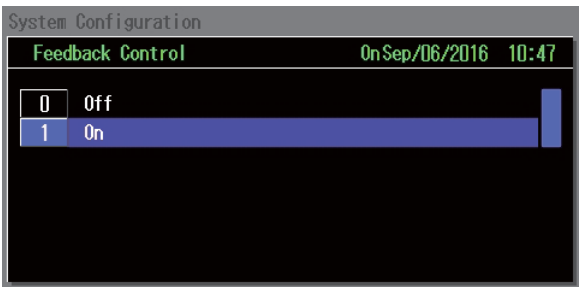


Fig. 10-61

10.5 FBC INDIRECT FEEDBACK CONTROL (PULSE OUTPUT)

This section provides information on the case DACS provides PULSE SIGNAL with the auger filler and the auger filler control part controls the auger filler in conjunction with the work of the auger filler and DACS.

10.5.1 Overview of the auger filler feedback control

This equipment controls the auger filler so that filler output achieves the Target Weight with the following structure.

DACS-GN checks the weight of the filled packs and judges whether the quantity is excessive or insufficient using the average value of the weight. Based on this, deviation volume against the Target Weight is converted into the pulse count or signal pulse width, which will be signal-output to the auger filler.

The auger filler is controlled by the control part of the auger filler.

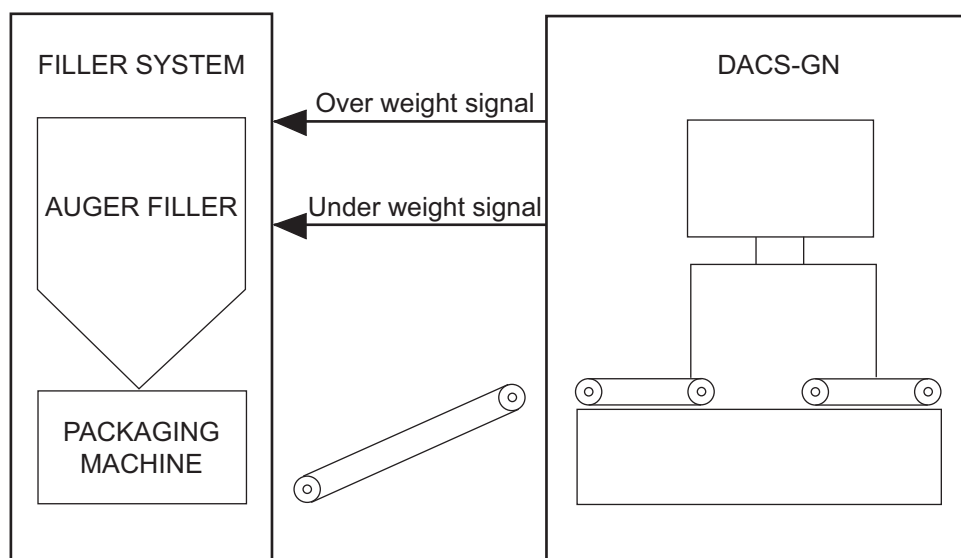


Fig. 10-62

10.5.2 Status Display

A display showing FBC status is added during the operation and standby. Pressing the [Display] key until the FBS status appears. Pressing the [Display] key each time switches the display.

Follow the following procedure to display the FBC Status Display.

1. Press [Display] key during operation/
standby.
- ▶Pressing [Display] key each time switches
the display in the following order:

"Weight Value Zoom Display"(Fig. 10-63)

↓

"Preset Information Display"(Fig. 10-64)

↓

"Histogram Display"(Fig. 10-65)

↓

"X-Bar Chart Display"(Fig. 10-66)

↓

"Last 20 Weight Data Display"(Fig. 10-67)

↓

"FBC READY STATUS SCREEN"
(Fig. 10-68)

↓

"Pitch Adjustment Display"(Fig. 10-69)

↓

"Total Count Zoom Display"(Fig. 10-70)

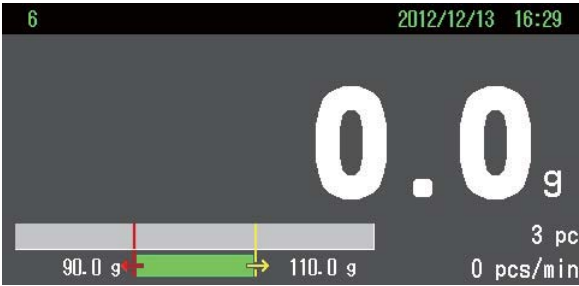


Fig. 10-63



Fig. 10-64



Fig. 10-65



Fig. 10-66



Fig. 10-67

NOTE

- On "FBC READY STATUS SCREEN" ("Fig. 10-68 "), items are indicated below.

for Sample Count A/B

B: The setting value of Sample Count

A: The remaining value of Sample Count for FBC

for Sample Interval C/D

D: The setting value of Sample Interval for FBC

C: The remaining value of Sample Interval before sampling start for FBC

- On "FBC READY STATUS SCREEN" ("Fig. 10-68 "), select "ON"/"OFF" by [Up/Down] key to turn the FBC ON/OFF.



Fig. 10-68



Fig. 10-69

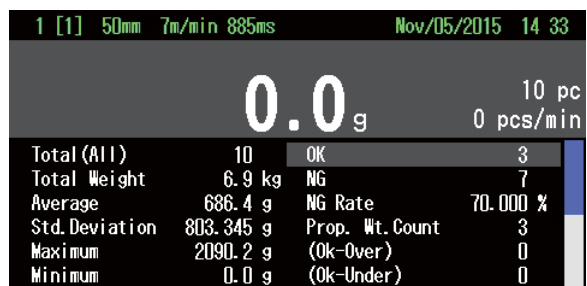


Fig. 10-70

10.5.3 Preset Parameters

An item for setting FBC is added to the Preset Setting. To access FBC Setting follow the procedures listed below.

NOTE

- Preset Setting is conducted with above level 1.

1. Display the Preset menu.
2. Select and enter the "Detailed Setting".
 - ▶ The Preset menu (Detailed) screen is displayed.
3. Select and enter the "FBC Setting".
 - ▶ The FBC Setting screen appears.

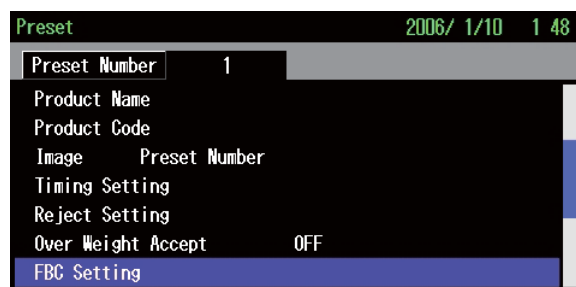


Fig. 10-71

10.5.3.1 FBC

This defines whether or not the equipment outputs the signal based on weighing results.

Setting "0" (OFF) does not output the signal.

Setting "1" (ON) outputs the signal.

To set FBC follow the procedures listed below.

1. Display the FBC Setting screen.
2. Select and enter the "FBC".

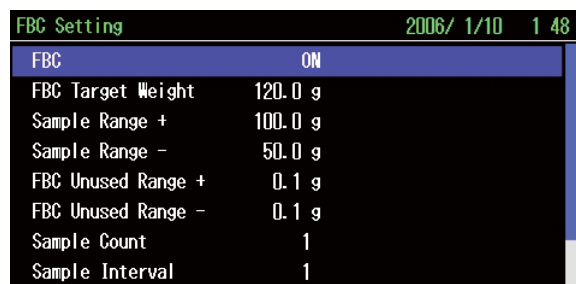


Fig. 10-72

3. Select and enter the "0" to deactivate Feedback Control.
Select and enter the "1" to activate Feedback Control.

▶ The FBC Setting screen is displayed.

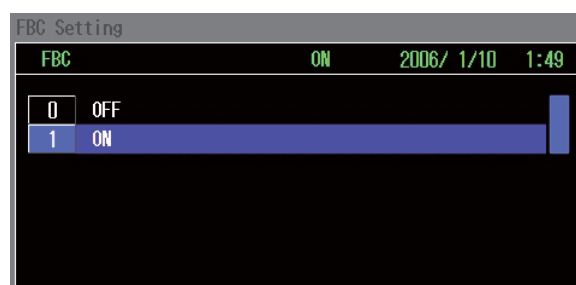


Fig. 10-73

10.5.3.2 FBC Target Weight

This defines the target net weight for packs filled by the auger filler.
To set FBC Target Weight follow the procedures listed below.

1. Display the FBC Setting screen.
2. Select and enter the "FBC Target Weight".

FBC Setting		2006/ 1/10	1:49
FBC	ON		
FBC Target Weight	120.0 g		
Sample Range +	100.0 g		
Sample Range -	50.0 g		
FBC Unused Range +	0.1 g		
FBC Unused Range -	0.1 g		
Sample Count	1		
Sample Interval	1		

Fig. 10-74

3. Input and enter the numeric value.
 - ▶ FBC Target Weight appears.
 - ▶ The setting for Target Weight is complete.

FBC Setting		2006/ 1/10	1:49
FBC Target Weight	120.0 g		
120.0			

Fig. 10-75

10.5.3.3 Sample Range +

This defines the maximum pack weight above which the pack will not be included in the sampling.
To set Sampling Range + follow the procedures listed below.

1. Display the FBC Setting screen.
2. Select and enter the "Sampling Range +".

FBC Setting		2006/ 1/10	1:49
FBC	ON		
FBC Target Weight	120.0 g		
Sample Range +	100.0 g		
Sample Range -	50.0 g		
FBC Unused Range +	0.1 g		
FBC Unused Range -	0.1 g		
Sample Count	1		
Sample Interval	1		

Fig. 10-76

- Input and enter the numeric value.

- ▶ Sampling Range + appears.
- ▶ The setting for Sampling Range + is complete.

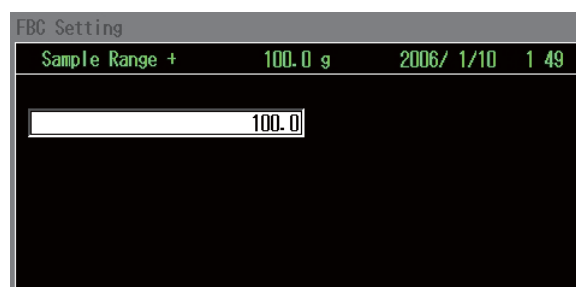


Fig. 10-77

10.5.3.4 Sample Range -

This defines the minimum pack weight below which the pack will not be included in the sampling. To set Sampling Range - follow the procedures listed below.

- Display the FBC Setting screen.
- Select and enter the "Sampling Range -".

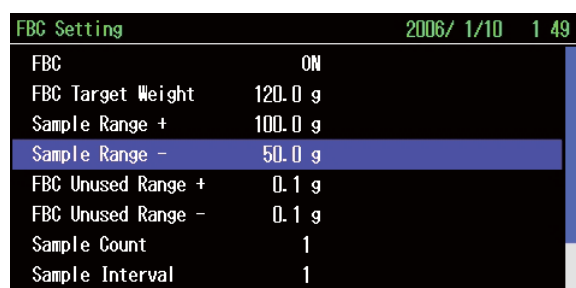


Fig. 10-78

- Input and enter the numeric value.

- ▶ Sampling Range - appears.
- ▶ The setting for Sampling Range - is complete.

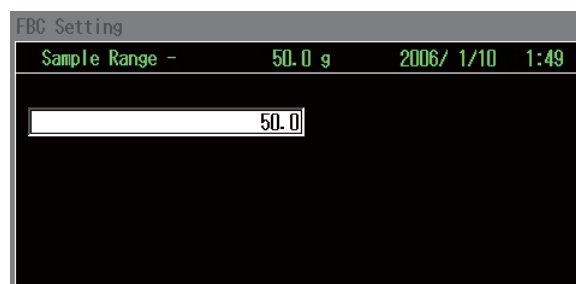


Fig. 10-79

10.5.3.5 FBC Unused Range +

This defines the average weight value which activates FBC when this weight is exceeded. To set FBC Unused Range + follow the procedures listed below.

1. Display the FBC Setting screen.
2. Select and enter the "FBC Unused Range +".

FBC Setting		2006/ 1/10 1:49
FBC	ON	
FBC Target Weight	120.0 g	
Sample Range +	100.0 g	
Sample Range -	50.0 g	
FBC Unused Range +	0.1 g	
FBC Unused Range -	0.1 g	
Sample Count	1	
Sample Interval	1	

Fig. 10-80

3. Input and enter the numeric value.
 - ▶ FBC Unused Range + appears.
 - ▶ The setting for FBC Unused Range + is complete.

FBC Setting		2006/ 1/10 1:49
FBC Unused Range +	0.1 g	
	0.1	

Fig. 10-81

10.5.3.6 FBC Unused Range -

This defines the average weight value below which activates FBC when this weight is not reached. To set FBC Unused Range - follow the procedures listed below.

1. Display the FBC Setting screen.
2. Select and enter the "FBC Unused Range -".

FBC Setting		2006/ 1/10 1:49
FBC	ON	
FBC Target Weight	120.0 g	
Sample Range +	100.0 g	
Sample Range -	50.0 g	
FBC Unused Range +	0.1 g	
FBC Unused Range -	0.1 g	
Sample Count	1	
Sample Interval	1	

Fig. 10-82

3. Input and enter the numeric value.
 - ▶ FBC Unused Range - appears.
 - ▶ The setting for FBC Unused Range - is complete.

FBC Setting		2006/ 1/10 1:49
FBC Unused Range -	0.1 g	
	0.1	

Fig. 10-83

10.5.3.7 Sample Count

This defines the number of packs from which to derive the average value.
To set Sample Count follow the procedures listed below.

- 1. Display the FBC Setting screen.
- 2. Select and enter the "Sample Count".

FBC Setting		2006/ 1/10	1 49
FBC	ON		
FBC Target Weight	120.0 g		
Sample Range +	100.0 g		
Sample Range -	50.0 g		
FBC Unused Range +	0.1 g		
FBC Unused Range -	0.1 g		
Sample Count	1		
Sample Interval	1		

Fig. 10-84

- 3. Input and enter the numeric value.
 - ▶ Sample Count appears.
 - ▶ The setting for Sample Count is complete.

NOTE

- Set the value 1 or above.

FBC Setting			
Sample Count	1	2006/ 1/10	1:49
1			

Fig. 10-85

10.5.3.8 Sample Interval

This defines the interval from when the first pack using a new pulse count is sent until it is check weighted by DACS. Sampling is not taken during this interval.
To set Sample Interval follow the procedures listed below.

- 1. Display the FBC Setting screen.
- 2. Select and enter the "Sample Interval".

FBC Setting		2006/ 1/10	1 50
FBC	ON		
FBC Target Weight	120.0 g		
Sample Range +	100.0 g		
Sample Range -	50.0 g		
FBC Unused Range +	0.1 g		
FBC Unused Range -	0.1 g		
Sample Count	1		
Sample Interval	1		

Fig. 10-86

3. Input and enter the numeric value.

- ▶ Sample Interval appears.
- ▶ The setting for Sample Interval is complete.

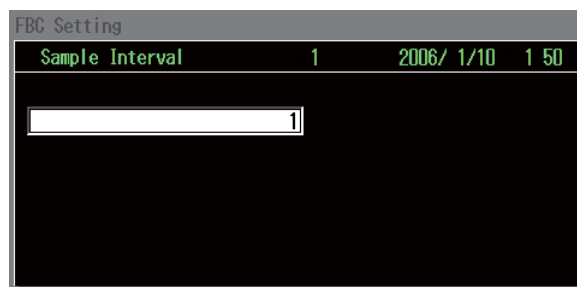


Fig. 10-87

10.5.3.9 FB Rate

This defines the ratio for converting the amount of FBC target weight and average sampled deviation into pulse signal output (count or width).

NOTE

- This value is different according to FB output type.
- For pulse count output:
Sets weight for one pulse.
Setting: [65.535 g/p] or less.
- For pulse width output:
Sets weight for one second.
Setting: [65.535 g/s] or less.

To set FB Rate follow the procedures listed below.

1. Display the FBC Setting screen.
2. Select and enter the "FB Rate".

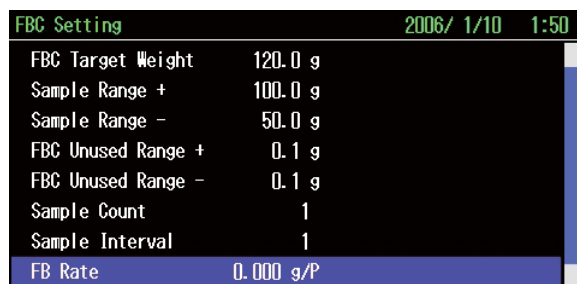


Fig. 10-88

3. Input and enter the numeric value.

- ▶ FB Rate appears.
- ▶ The setting for FB Rate is complete

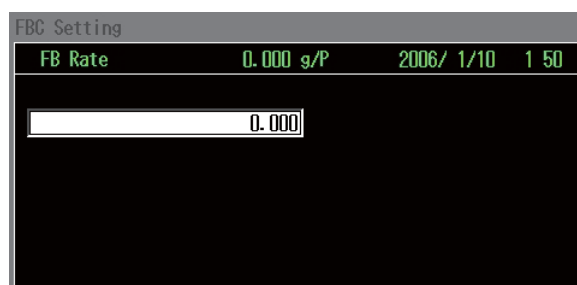


Fig. 10-89

10.5.3.10 Control Pulse Frequency

This item sets the frequency of output signal.

This item is only used when FB Output Mode is set to Pulse Count Output.

To set Control Pulse Frequency follow the procedures listed below.

1. Display the FBC Setting screen.
2. Select and enter the "Control Pulse Frequency".

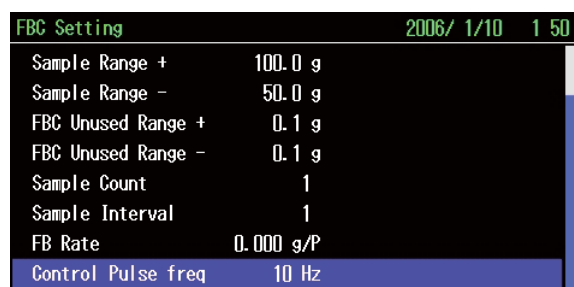


Fig. 10-90

3. Input and enter the numeric value.
 - ▶Control Pulse Frequency appears.
 - ▶The setting for Control Pulse freq is complete.

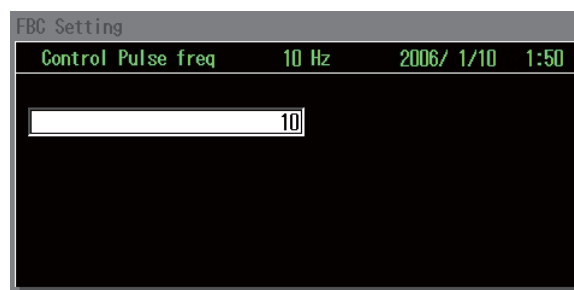


Fig. 10-91

NOTE

- Setting: 500 Hz or less.
- The control cycle of output signal is 10 ms. Accordingly, the frequency that is actually outputted will be rounded to the units of 10 ms.

10.5.4 System Configuration

An item is added to the System Configuration menu displaying FBC specifications.

Sets whether or not to control auger filler output with this equipment in conjunction with the auger filler. Also, if to control, this sets whether the amount of target weight and average sampled deviation is output by pulse count or pulse width.

To set FB Output Mode follow the procedures listed below.

NOTE

- System Configuration is conducted with above level 2.

1. Display the System Configuration menu.
2. Select and enter the "FB Output Mode".
 ► The FB Output Mode display appears.

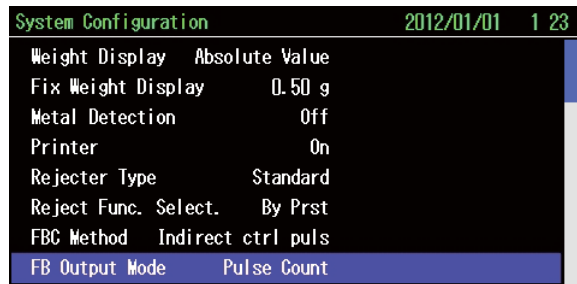


Fig. 10-92

3. Select and enter the "0" not to control auger filler output.
 Select and enter the "1" to control auger filler output by outputting the deviation with pulse count,
 Select and enter the "2" to control auger filler output by outputting the deviation with pulse width.

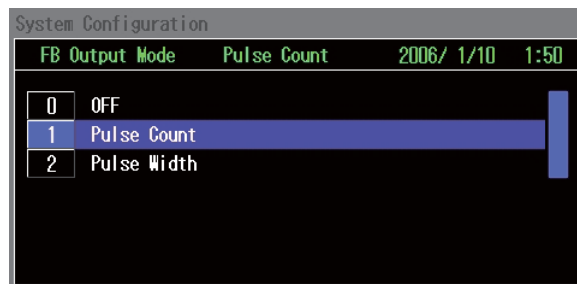


Fig. 10-93

- System Configuration display in which FB Output Mode is set appears.

10.6 FBC INDIRECT FEEDBACK CONTROL (CONTROL AMOUNT OUTPUT)

This section provides information on the case DACS provides CONTROL AMOUNT with the auger filler and the auger filler control part controls the auger filler in conjunction with the work of the auger filler and DACS.

10.6.1 Overview of the auger filler feedback control

This equipment controls the auger filler so that filler output achieves the Target Weight with the following structure.

DACS-GN checks the weight of the filled packs and judges whether the quantity is excessive or insufficient using the average value of the weight. Based on this, deviation volume against the Target Weight is converted into the Control Amount, which will be signal-output to the auger filler.

The auger filler is controlled by the control part of the auger filler.

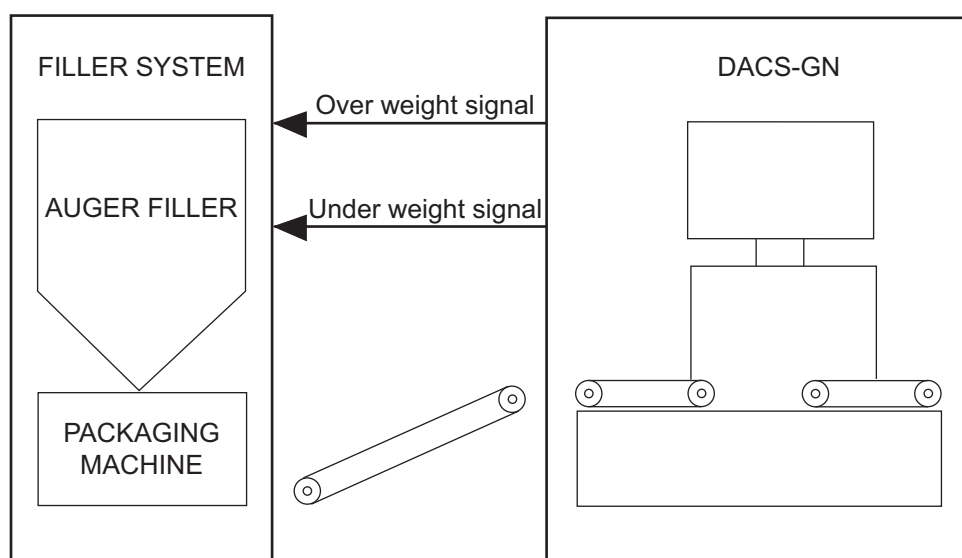


Fig. 10-94

10.6.2 Status Display

A display showing FBC status is added during the operation and standby. Pressing the [Display] key until the FBS status appears. Pressing the [Display] key each time switches the display. Follow the following procedure to display the FBC Status Display.

1. Press [Display] key during operation/standby.

► Pressing [Display] key each time switches the display in the following order:
"Weight Value Zoom Display"(Fig. 10-95)

↓
"Preset Information Display"(Fig. 10-96)

↓
"Histogram Display"(Fig. 10-96)

↓
"X-Bar Chart Display"(Fig. 10-98)

↓
"Last 20 Weight Data Display"(Fig. 10-99)

↓
"FBC READY STATUS SCREEN"(Fig. 10-100)

↓
"Pitch Adjustment Display"(Fig. 10-100)

↓
"Total Count Zoom Display"(Fig. 10-102)

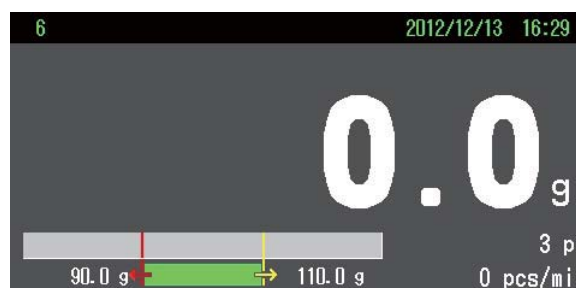


Fig. 10-95



Fig. 10-96



Fig. 10-97



Fig. 10-98

TIP

- On "FBC READY STATUS SCREEN" ("Fig. 10-100"), items are indicated below.

for Sample Count A/B

B: The setting value of Sample Count

A: The remaining value of Sample Count for FBC for Sample Interval C/D

D: The setting value of Sample Interval for FBC

C: The remaining value of Sample Interval before sampling start for FBC

- On "FBC READY STATUS SCREEN" ("Fig. 10-100"), select "ON"/"OFF" by [Up/Down] key to turn the FBC ON/OFF.



Fig. 10-99



Fig. 10-100



Fig. 10-101

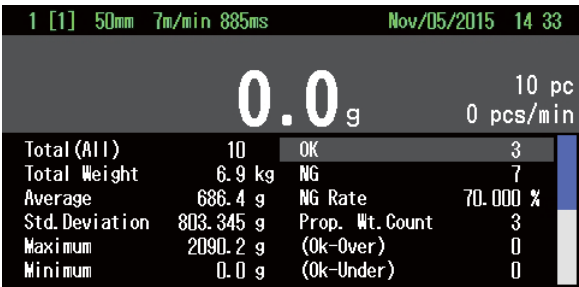


Fig. 10-102

10.6.3 Preset Parameters

An item for setting FBC is added to the Preset Setting. To access FBC Setting follow the procedures listed below.

NOTE

- Preset Setting is conducted with above level 1.

1. Display the Preset Setting screen.
2. Select and enter the "Detailed Setting".
 - ▶The Preset menu (Detailed) screen is displayed.
3. Select and enter the "FBC Setting".
 - ▶The FBC Setting screen is displayed.



Fig. 10-103

10.6.3.1 FBC

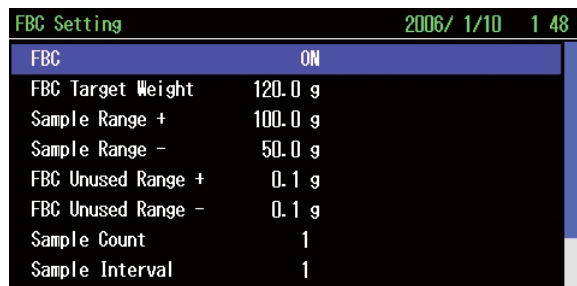
This defines whether or not the equipment outputs the signal based on weighing results.

Setting "0" (OFF) does not output the signal.

Setting "1" (ON) outputs the signal.

To set FBC follow the procedures listed below.

1. Display the FBC Setting screen.
2. Select and enter the "FBC".

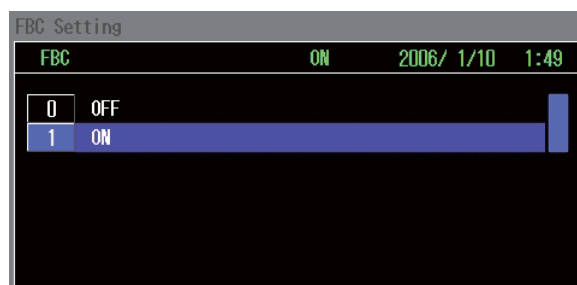


FBC Setting		2006/ 1/10 1 48
FBC	ON	
FBC Target Weight	120.0 g	
Sample Range +	100.0 g	
Sample Range -	50.0 g	
FBC Unused Range +	0.1 g	
FBC Unused Range -	0.1 g	
Sample Count	1	
Sample Interval	1	

Fig. 10-104

3. Select and enter the "0" to deactivate Feedback Control.
Select and enter the "1" to activate Feedback Control.

►The FBC Setting screen appears.



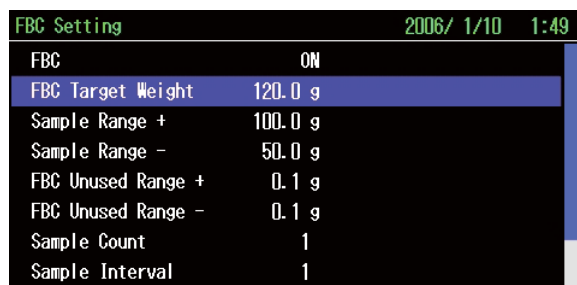
FBC Setting		2006/ 1/10 1:49
FBC	ON	
0	OFF	
1	ON	

Fig. 10-105

10.6.3.2 FBC Target Weight

This defines the target net weight for packs filled by the auger filler.
To set FBC Target Weight follow the procedures listed below.

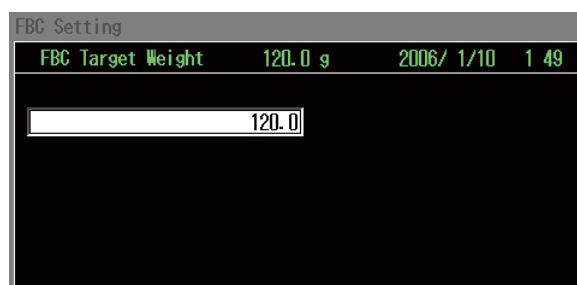
1. Display the FBC Setting screen.
2. Select and enter the "FBC Target Weight".



FBC Setting		2006/ 1/10 1:49
FBC	ON	
FBC Target Weight	120.0 g	
Sample Range +	100.0 g	
Sample Range -	50.0 g	
FBC Unused Range +	0.1 g	
FBC Unused Range -	0.1 g	
Sample Count	1	
Sample Interval	1	

Fig. 10-106

3. Input and enter the numeric value.
►FBC Target Weight appears.
►The setting for Target Weight is complete.



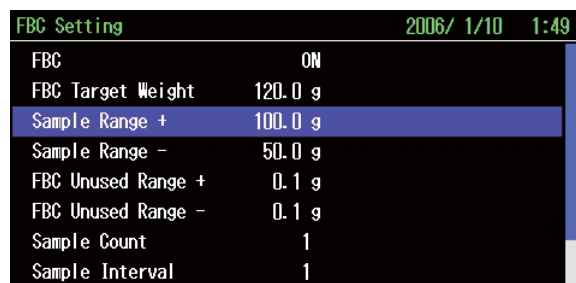
FBC Setting		2006/ 1/10 1 49
FBC Target Weight	120.0 g	
120.0		

Fig. 10-107

10.6.3.3 Sample Range +

This defines the maximum pack weight above which the pack will not be included in the sampling. To set Sampling Range + follow the procedures listed below.

1. Display the FBC Setting screen.
2. Select and enter the "Sampling Range +".



FBC Setting		2006/ 1/10	1:49
FBC	ON		
FBC Target Weight	120.0 g		
Sample Range +	100.0 g		
Sample Range -	50.0 g		
FBC Unused Range +	0.1 g		
FBC Unused Range -	0.1 g		
Sample Count	1		
Sample Interval	1		

Fig. 10-108

3. Input and enter the numeric value.
 - ▶ Sampling Range + appears.
 - ▶ The setting for Sampling Range + is complete.

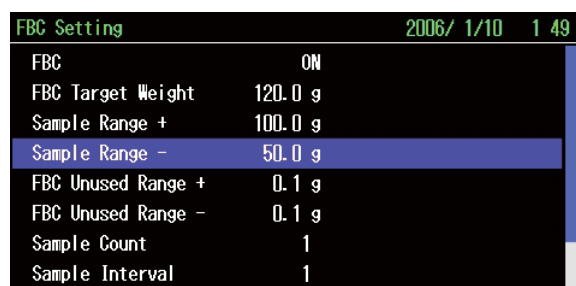
FBC Setting			
Sample Range +	100.0 g	2006/ 1/10	1 49
100.0			

Fig. 10-109

10.6.3.4 Sample Range -

This defines the minimum pack weight below which the pack will not be included in the sampling. To set Sampling Range - follow the procedures listed below.

1. Display the FBC Setting screen.
2. Select and enter the "Sampling Range -".



FBC Setting		2006/ 1/10	1:49
FBC	ON		
FBC Target Weight	120.0 g		
Sample Range +	100.0 g		
Sample Range -	50.0 g		
FBC Unused Range +	0.1 g		
FBC Unused Range -	0.1 g		
Sample Count	1		
Sample Interval	1		

Fig. 10-110

3. Input and enter the numeric value.

- ▶ Sampling Range - appears.
- ▶ The setting for Sampling Range - is complete.

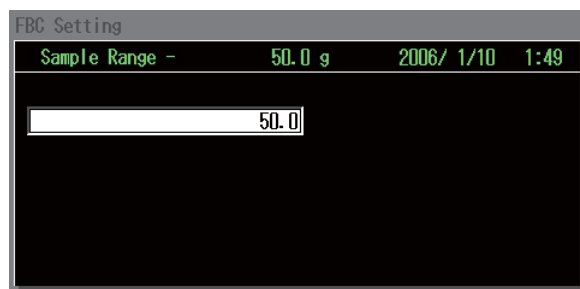


Fig. 10-111

10.6.3.5 FBC Unused Range +

This defines the average weight value which activates FBC when this weight is exceeded. To set FBC Unused Range + follow the procedures listed below.

1. Display the FBC Setting screen.
2. Select and enter the "FBC Unused Range +".

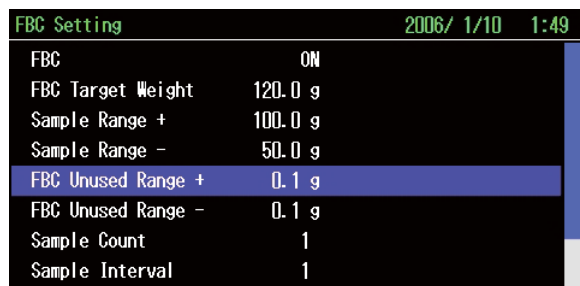


Fig. 10-112

3. Input and enter the numeric value.

- ▶ FBC Unused Range + appears.
- ▶ The setting for FBC Unused Range + is complete.

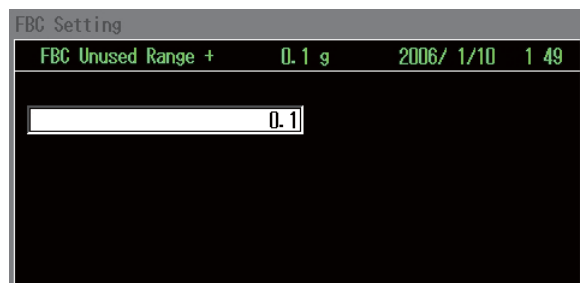
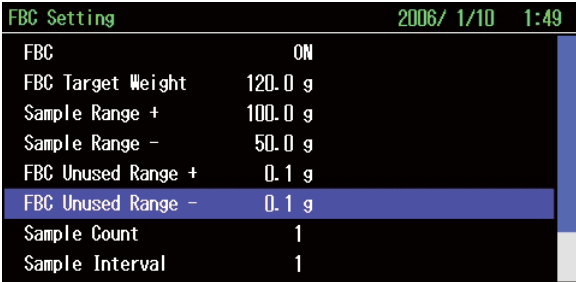


Fig. 10-113

10.6.3.6 FBC Unused Range -

This defines the average weight value below which activates FBC when this weight is not reached. To set FBC Unused Range - follow the procedures listed below.

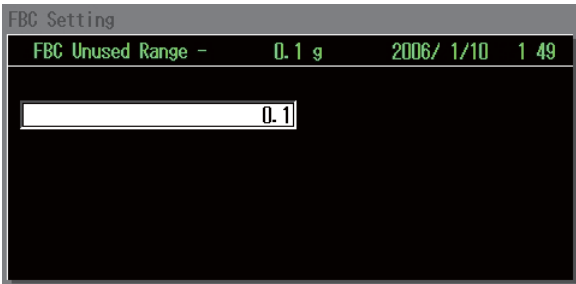
1. Display the FBC Setting screen.
2. Select and enter the "FBC Unused Range -".



FBC Setting		2006/ 1/10	1:49
FBC	ON		
FBC Target Weight	120.0 g		
Sample Range +	100.0 g		
Sample Range -	50.0 g		
FBC Unused Range +	0.1 g		
FBC Unused Range -	0.1 g		
Sample Count	1		
Sample Interval	1		

Fig. 10-114

3. Input and enter the numeric value.
 - ▶ FBC Unused Range - appears.
 - ▶ The setting for FBC Unused Range - is complete.



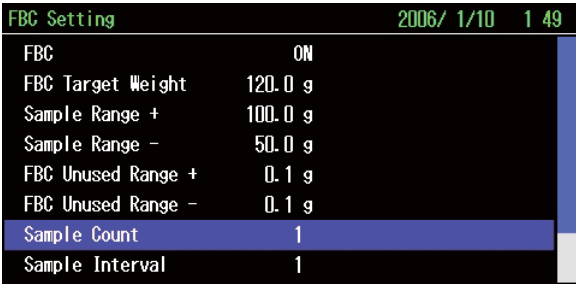
FBC Setting		2006/ 1/10	1:49
FBC Unused Range -	0.1 g		
0.1			

Fig. 10-115

10.6.3.7 Sample Count

This defines the number of packs from which to derive the average value.
To set Sample Count follow the procedures listed below.

1. Display the FBC Setting screen.
2. Select and enter the "Sample Count".



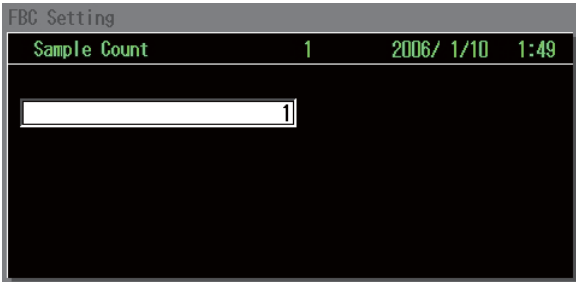
FBC Setting		2006/ 1/10	1:49
FBC	ON		
FBC Target Weight	120.0 g		
Sample Range +	100.0 g		
Sample Range -	50.0 g		
FBC Unused Range +	0.1 g		
FBC Unused Range -	0.1 g		
Sample Count	1		
Sample Interval	1		

Fig. 10-116

3. Input and enter the numeric value.
 - ▶ Sample Count appears.
 - ▶ The setting for Sample Count is complete.

NOTE

- Set the value 1 or above.



FBC Setting		2006/ 1/10	1:49
Sample Count	1		
1			

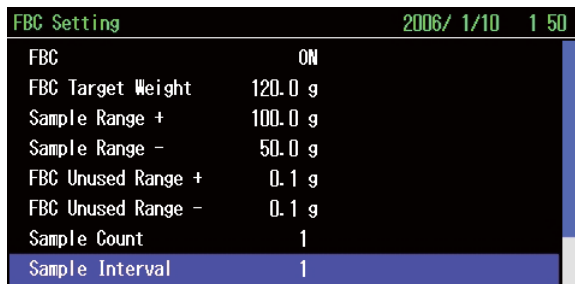
Fig. 10-117

10.6.3.8 Sample Interval

This defines the interval from when the first pack using a new pulse count is sent until it is checked by DACS. Sampling is not taken during this interval.

To set Sample Interval follow the procedures listed below.

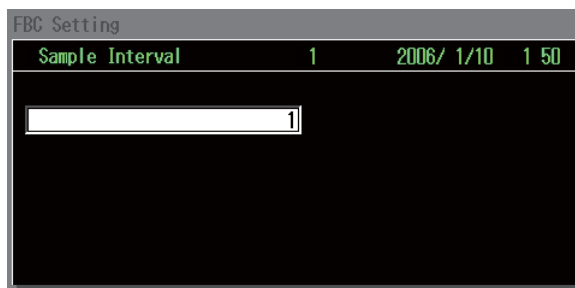
1. Display the FBC Setting screen.
2. Select and enter the "Sample Interval".



FBC Setting		2006/ 1/10 1 50
FBC	ON	
FBC Target Weight	120.0 g	
Sample Range +	100.0 g	
Sample Range -	50.0 g	
FBC Unused Range +	0.1 g	
FBC Unused Range -	0.1 g	
Sample Count	1	
Sample Interval	1	

Fig. 10-118

3. Input and enter the numeric value.
 - ▶ Sample Interval appears.
 - ▶ The setting for Sample Interval is complete.



FBC Setting		2006/ 1/10 1 50
Sample Interval	1	
1		

Fig. 10-119

10.6.4 System Configuration

An item is added to the System Configuration menu displaying FBC specifications.

Sets whether or not to control auger filler output with this equipment in conjunction with the auger filler. Also, if to control, this sets whether or not the amount of target weight and average sampled deviation is output.

To activate Feedback Control follow the procedures listed below.

NOTE

- System Configuration is conducted with above level 2.

1. Display the System Configuration menu.
2. Select and enter the "Feedback Control".

► The Feedback Control display appears.

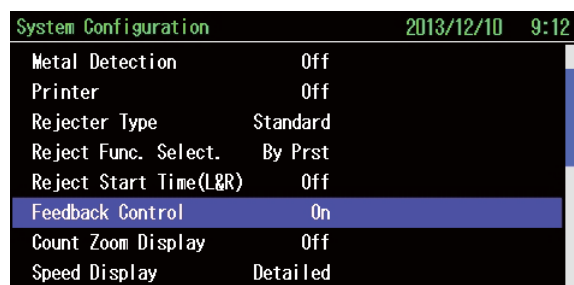


Fig. 10-120

3. Select and enter the "0" to deactivate Feedback Control.
Select and enter the "1" to activate Feedback Control.

► System Configuration display in which Feedback Control is set appears.

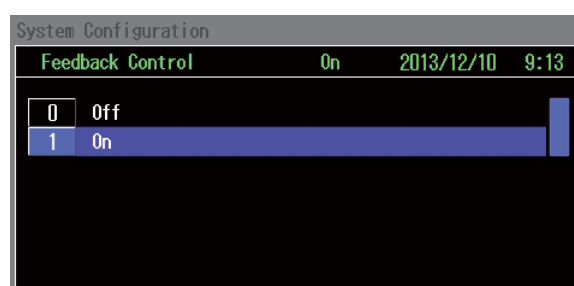


Fig. 10-121

10.7 FLOATING REF. WT. REVISION

This section provides information on the Ref. Wt. update specification.

NOTE

- If the weight zone is set to "5 Zone", FLOATING REF. WT. REVISION cannot be used.

10.7.1 Overview

Floating Reference Weight Revision functions as follows. When fluctuations in the average weight of product are relatively large, the reference weight is revised to equal the average weight of each pack as it is weighed. In this way, weight variations do not adversely affect the weighing operation. This is especially effective in checking for missing pieces.

EXAMPLE: If the Reference Weight Revision Piece Count is 3, the Reference Weight value will be revised in the following manner:

1. **PROPER WT. PACK NUMBERS**
(Pack Number 1 is the 1st proper weight product to be checkweighed for descriptive purpose.)
2. Average weight of proper weight Pack Numbers 1, 2 and 3 will be used as the Reference Weight in the next weighing.
3. Average weight of proper weight Pack Numbers 2, 3, and 4 will be used as the Reference Weight in next weighing.
4. Average weight of proper weight Pack Numbers 3, 4, and 5 will be used as the Reference Weight in the next weighing.

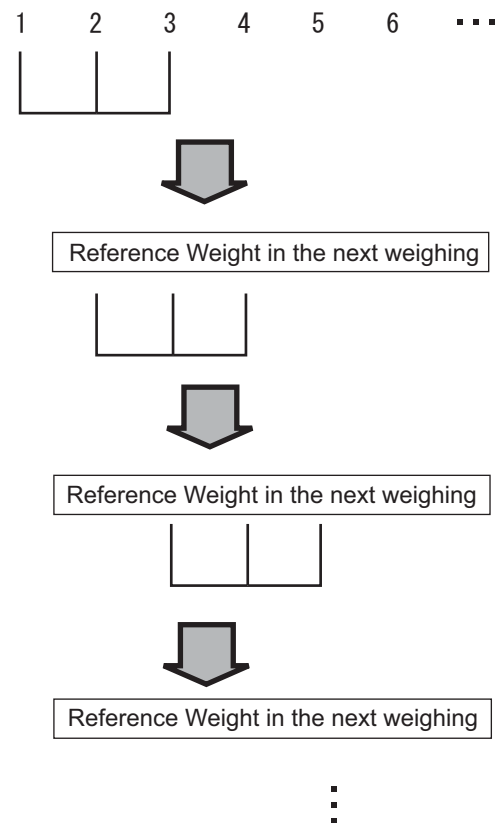


Fig. 10-122

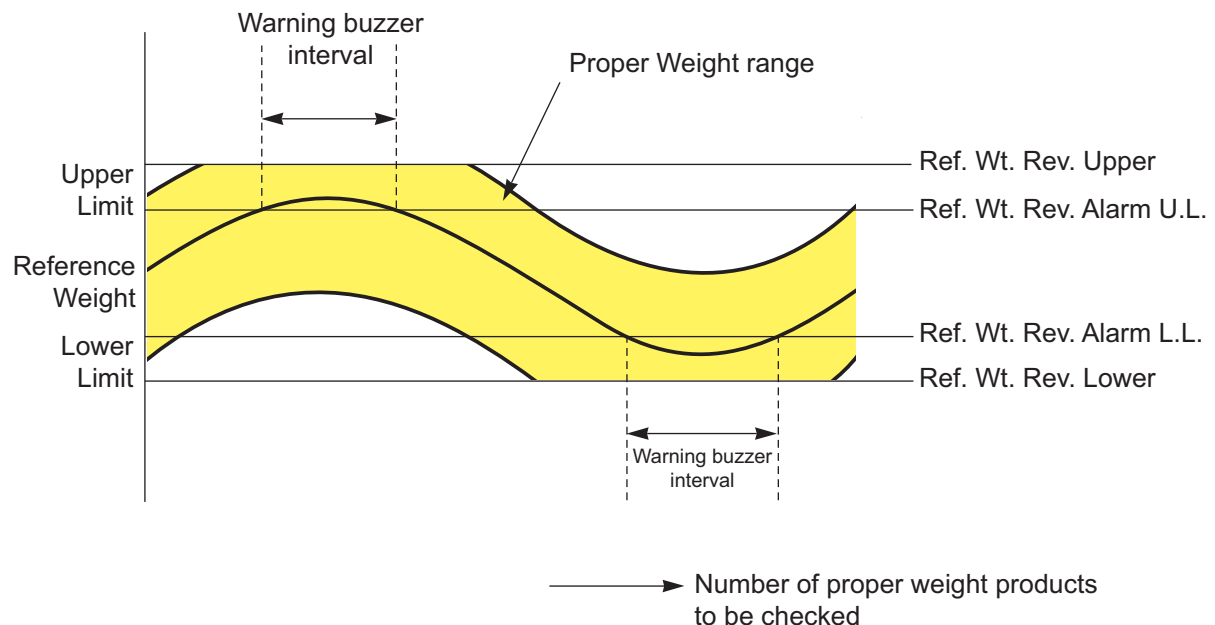


Fig. 10-123

As the reference weight value is revised, the upper and lower limits are revised accordingly. However, the upper limit value never exceeds the reference weight revision lower limit value. In other words, any product pack which exceeds the reference weight revision upper limit is judged as over weight, and any product pack which weighs less than the reference weight revision lower limit is judged as under weight. If the reference weight exceeds the reference weight alarm upper limit, "Upper limit alarm" appears and a buzzer sounds.

If the reference weight is less than the reference weight alarm lower limit, "Lower limit alarm" and a buzzer sounds.

10.7.2 Preset Parameters

Five Preset Parameter settings concerning FLOATING REF. WT. REVISION are added.

NOTE

- Preset Setting is conducted with above level 1.
- Displayed when Setting "ON" for the FLOATING REF. WT. REVISION in the System Configuration. (Refer to "10.7.3 System Configuration")

10.7.2.1 Ref. Wt. Rev. Upper

When [Ref. Wt. + UPPER LIMIT] exceeds this value, this value becomes the upper limit, and packs exceeding this value are judged over weight.

To set Ref. Wt. Rev. Upper follow the procedures listed below.

1. Display the Preset menu.
2. Select and enter the "Detailed Setting".
 - ▶ The Preset menu (Detailed) screen is displayed.
3. Select and enter the "Ref. Wt. Rev. Upper".

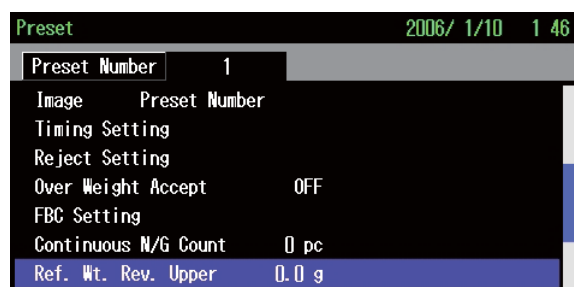


Fig. 10-124

4. Input and enter the numeric value.
 - ▶ Ref. Wt. Rev. Upper appears.
 - ▶ The setting for Ref. Wt. Rev. Upper is complete.

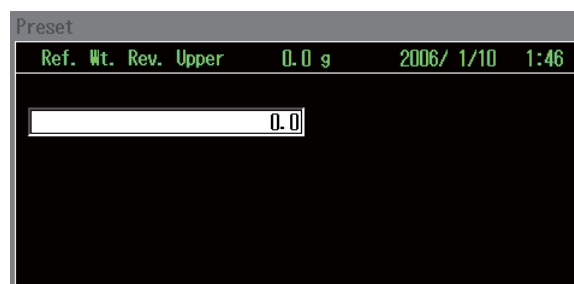


Fig. 10-125

NOTE

- This parameter is entered as an absolute value.

10.7.2.2 Ref. Wt. Rev. Lower

When Ref. Wt. Rev. Lower is less than this value, this value becomes the lower limit, and packs weighing less than this value are judged under weight.

To set Ref. Wt. Rev. Lower follow the procedures listed below.

1. Display the Preset menu.
2. Select and enter the "Detailed Setting".
 - ▶ The Preset menu (Detailed) screen is displayed.
3. Select and enter the "Ref. Wt. Rev. Lower".

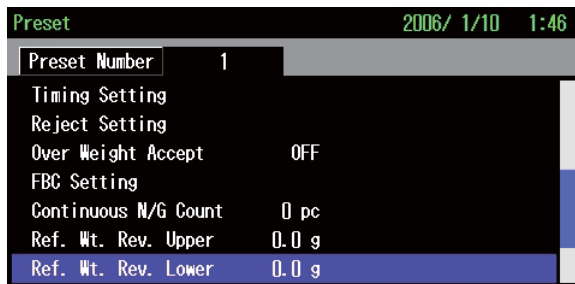


Fig. 10-126

4. Input and enter the numeric value.
 - ▶ Ref. Wt. Rev. Lower appears.
 - ▶ The setting for Ref. Wt. Rev. Lower is complete.

NOTE

- This parameter is entered as an absolute value.

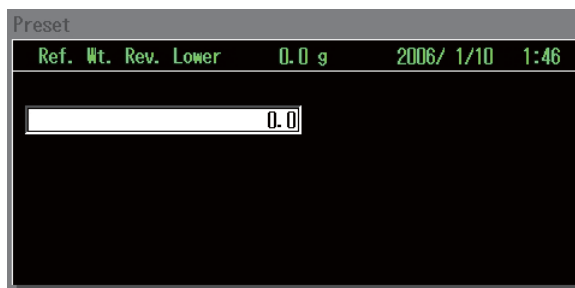


Fig. 10-127

10.7.2.3 Ref. Wt. Rev. Count

When Ref. Wt. Rev. is ON, proper weight data is sampled and reference weight is revised based on the average weight of one product. This item sets the count which is used to calculate the average weight. To set Ref. Wt. Rev. Count follow the procedures listed below.

1. Display the Preset menu.
2. Select and enter the "Detailed Setting".
 - ▶ The Preset menu (Detailed) screen is displayed.
3. Select and enter the "Ref. Wt. Rev. Count".

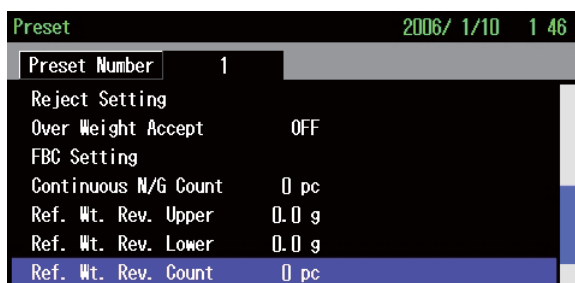


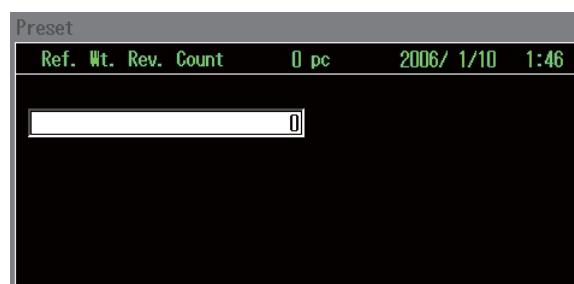
Fig. 10-128

4. Input and enter the numeric value.

- ▶ Ref. Wt. Rev. Count appears.
- ▶ The setting for Ref. Wt. Rev. Count is complete.

NOTE

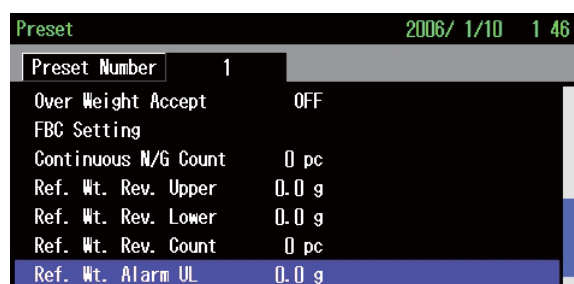
- Setting Range: 1 to 99.
- If "0" is set, reference weight revision is not performed. However, the Ref. Wt. Rev. Upper and Lower Limit values will still be in effect.

**Fig. 10-129****10.7.2.4 Ref. Wt. Rev. Alarm U.L.**

If Ref. Wt. Rev. is ON and when the reference weight value is exceeded, the Upper Limit Alarm message will be displayed and a warning buzzer will sound.

To set Ref. Wt. Rev. Alarm U.L. follow the procedures listed below.

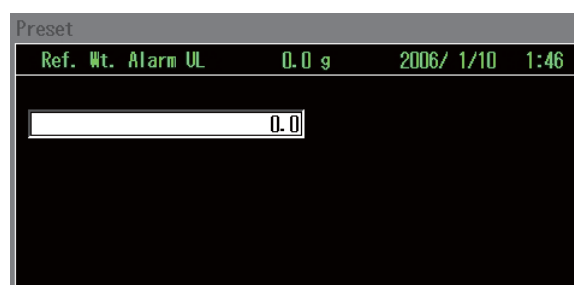
1. Display the Preset menu.
2. Select and enter the "Detailed Setting".
 - ▶ The Preset menu (Detailed) screen is displayed.
3. Select and enter the "Ref. Wt. Rev. Alarm U.L.".

**Fig. 10-130**

4. Input and enter the numeric value.
 - ▶ Ref. Wt. Rev. Alarm U.L. appears.
 - ▶ The setting for Ref. Wt. Rev. Alarm U.L. is complete.

NOTE

- This parameter is entered as an absolute value.

**Fig. 10-131**

10.7.2.5 Ref. Wt. Rev. Alarm L.L.

If Ref. Wt. Rev. is ON and when the reference weight is less than this value, the Lower Limit Alarm message will be displayed and a warning buzzer will sound.

To set Ref. Wt. Rev. Alarm L.L. follow the procedures listed below.

1. Display the Preset menu.
2. Select and enter the "Detailed Setting".
 - ▶ The Preset menu (Detailed) screen is displayed.
3. Select and enter the "Ref. Wt. Rev. Alarm L.L.".
 - ▶ The Preset menu (Detailed) screen is displayed.

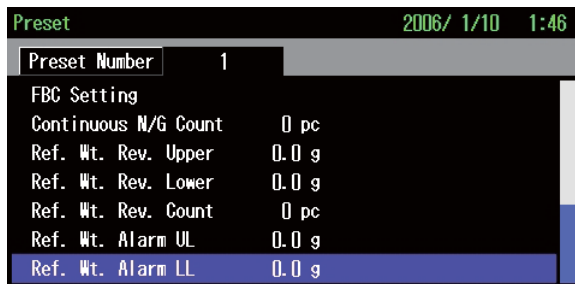


Fig. 10-132

4. Input and enter the numeric value.
 - ▶ Ref. Wt. Rev. Lower appears.
 - ▶ The setting for Ref. Wt. Rev. Lower is complete.

NOTE

- This parameter is entered as an absolute value.

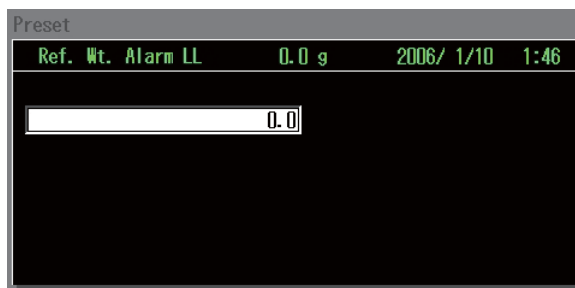


Fig. 10-133

10.7.3 System Configuration

An item is added to the System Configuration menu displaying Ref. Wt. Rev.

To initiate Ref. Wt. Rev. follow the procedures listed below:

NOTE

- System Configuration is conducted with above level 2.

1. Display the System Configuration menu.
2. Select and enter the "Ref. Wt. Rev."
 - ▶Ref. Wt. Rev. display appears.

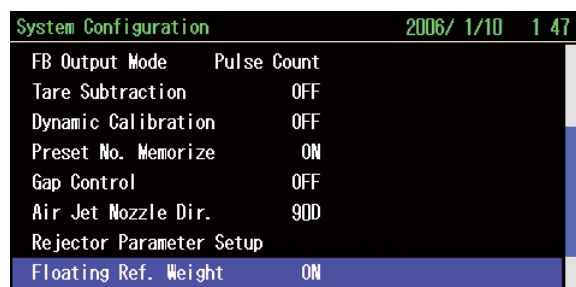


Fig. 10-134

3. Select and enter the "0" to deactivate Ref. Wt. Rev.
Select and enter the "1" to activate Ref. Wt. Rev.
 - ▶System Configuration display in which Ref. Wt. Rev. is set appears.

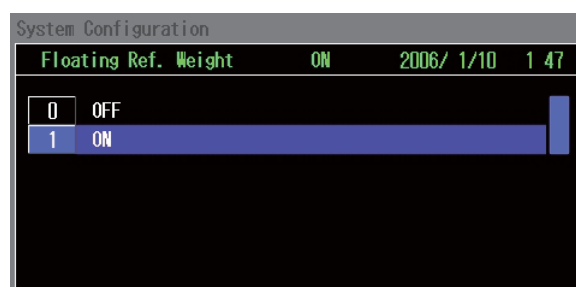


Fig. 10-135

10.8 Weight Data Output

This section describes the specifications of weight output when weighing.

1. The communications standards below are supported.
 - RS-232C
 - Ethernet
2. The output formats of weight data (standard) are as follows:

Weigh Data Output			
0)	Control Code	LF (0Ah)	
1)	Command	'G' (47h)	
2)	Weight Status		
	Proper	'1' (31h)	
	Under	'2' (32h)	
	Over	'3' (33h)	
	Metal	'4' (34h)	
	Ext.1	'5' (35h)	
	Ext.2	'6' (36h)	
	Pitch Error	'7' (37h)	
	0-error	'8' (38h)	
	Cont.NG	'9' (39h)	
	No Metal	'.' (3Ah)	
	OK-under	';' (3Bh)	
	OK-over	'<' (3Ch)	
	Under-NG	'=' (3Dh)	
	Over-NG	'>' (3Eh)	
	Foreign Obj	'@' (40h)	
	Shorter Item	'A' (41h)	
	Product Length Error	'B' (42h)	
	Ext.3	'C' (43h)	
	Ext.4	'D' (44h)	
3)	Weight Data	'0'(30h) - '9'(39h)	OR '-' (2Dh)
4)	Weight Data	'0'(30h) - '9'(39h)	
5)	Weight Data	'0'(30h) - '9'(39h)	OR '.' (2Eh)
6)	Weight Data	'0'(30h) - '9'(39h)	OR '.' (2Eh)
7)	Weight Data	'0'(30h) - '9'(39h)	OR '.' (2Eh)
8)	Weight Data	'0'(30h) - '9'(39h)	OR '.' (2Eh)
9)	Weight Data	'0'(30h) - '9'(39h)	OR '.' (2Eh)
10)	Weight Data	'0'(30h) - '9'(39h)	
11)	Weight Unit	'g'(67h: gram)	OR 'o'(6Fh: ounce)
12)	Checksum	'0'(30h) - '?'(3Fh)	
13)	Checksum	'0'(30h) - '?'(3Fh)	
14)	Control Code	CR (0Dh)	

Fig. 10-136

The weigh data is output only once after weighing.

If the weighing result is went off the scale, 3) to 10) above display "-"(2Dh).

Output target of weighing result is all the status, and the result is output every time after weighing.

The output weight is absolute value or deviation value in conjunction with the display of DACS-GN.

3. The output formats of weight data (special) are as follows:

Weigh Data Output			
0)	Control Code	LF (0Ah)	
1)	Command	'G' (47h)	
2)	Machine No.	'0'(30h) - '9'(39h)	
3)	Machine No.	'0'(30h) - '9'(39h)	
4)	Machine No.	'0'(30h) - '9'(39h)	
5)	Machine No.	'0'(30h) - '9'(39h)	
6)	Machine No.	'0'(30h) - '9'(39h)	
7)	Preset No.	'0'(30h) - '9'(39h)	
8)	Preset No.	'0'(30h) - '9'(39h)	
9)	Preset No.	'0'(30h) - '9'(39h)	
10)	Weight Status		
	Proper	'1' (31h)	
	Under	'2' (32h)	
	Over	'3' (33h)	
	Metal	'4' (34h)	
	Ext.1	'5' (35h)	
	Ext.2	'6' (36h)	
	Pitch Error	'7' (37h)	
	0-error	'8' (38h)	
	Cont.NG	'9' (39h)	
	No Metal	'.' (3Ah)	
	OK-under	';' (3Bh)	
	OK-over	'<' (3Ch)	
	Under-NG	'=' (3Dh)	
	Over-NG	'>' (3Eh)	
	Foreign Obj	'@' (40h)	
	Shorter Item	'A' (41h)	
	Product Length Error	'B' (42h)	
	Ext.3	'C' (43h)	
	Ext.4	'D' (44h)	
11)	Weight Data	'0'(30h) - '9'(39h)	OR '-' (2Dh)
12)	Weight Data	'0'(30h) - '9'(39h)	
13)	Weight Data	'0'(30h) - '9'(39h)	OR '.' (2Eh)
14)	Weight Data	'0'(30h) - '9'(39h)	OR '.' (2Eh)
15)	Weight Data	'0'(30h) - '9'(39h)	OR '.' (2Eh)
16)	Weight Data	'0'(30h) - '9'(39h)	OR '.' (2Eh)
17)	Weight Data	'0'(30h) - '9'(39h)	OR '.' (2Eh)
18)	Weight Data	'0'(30h) - '9'(39h)	
19)	Weight Unit	'g'(67h: gram)	OR 'o'(6Fh: ounce)
20)	Checksum	'0'(30h) - '?'(3Fh)	
21)	Checksum	'0'(30h) - '?'(3Fh)	
22)	Control Code	CR (0Dh)	

Fig. 10-137

The weigh data is output only once after weighing.

If the weighing result is went off the scale, 3) to 10) above display "-"(2Dh).

Output target of weighing result is all the status, and the result is output every time after weighing.

The output weight is absolute value or deviation value in conjunction with the display of DACS-G.N

